

Dreaming rich: How inequality manufactures overoptimism

Julia Baumann¹ Yiming Liu²

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Abstract

The prospect of upward mobility (POUM) hypothesis holds that high inequality can be sustained when combined with high mobility: people may oppose redistribution because they expect to be richer tomorrow. In reality, however, high inequality is associated with low mobility. We propose and test a novel mechanism that resolves this tension: higher inequality causes motivated optimism about upward mobility. In survey data from 27 countries, higher inequality is associated with greater expected upward mobility despite lower actual mobility, both across countries and individuals. Higher mobility beliefs are in turn associated with weaker redistribution demand, consistent with POUM. We then test the mechanism in an online experiment that randomly assigns participants to a society with high or low inequality. Higher inequality raises overoptimism about reaching the high-paying position, while lowering beliefs about positions above the median that pay below mean income, a pattern consistent with motivated beliefs and robust to accurate performance feedback. This induced overoptimism causally raises the inequality participants accept when they choose the income distribution. We conclude with the welfare implications of this motivated optimism, weighing the hedonic utility it provides against how it sustains inequality.

Keywords: Income inequality, Inequality acceptance, Motivated beliefs, Overconfidence

JEL Codes: D31, D63, D83, H24

¹Humboldt-Universität zu Berlin and WZB; julia.baumann@wzb.eu

²Humboldt-Universität zu Berlin and WZB; yiming.liu@hu-berlin.de

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Ever since De Tocqueville (1835), a prominent explanation for why democratic societies tolerate high inequality is that citizens believe in upward mobility (Benabou and Ok, 2001; Alesina et al., 2018). If people expect that they or their children will move up the income ladder, they are more willing to accept large income gaps today. This prospect of upward mobility (POUM) logic implies a positive relationship between inequality and mobility: the societies that sustain the most inequality should be those where people are most likely to move up. The data point the other way. Using data from the International Social Survey Programme (ISSP), which covers 27 countries and records each respondent’s social status, their parents’ status, and their expected future status, we measure realized mobility as the share of people whose status exceeds their parents’. Consistent with the Great Gatsby Curve (Corak, 2013), this share is *lower* in more unequal countries (Panel (a) of Figure 1), the opposite of what POUM appears to require. We reconcile this conflict with a distinction at the core of POUM: it requires only that people *believe* they will move up, not that they actually do. We propose and test a novel psychological mechanism through which high inequality itself produces this belief: by widening the gap between the top and the average, it raises the hedonic benefit of believing one will reach the top, motivating over-optimism about getting there.

Figure 1 consists of two scatter plots, (a) and (b), showing the relationship between the Gini coefficient (x-axis) and mobility (y-axis). Both plots include a dashed regression line indicating a negative correlation.

(a) Gini and actual mobility

The y-axis is labeled "Actual upward mobility (share)" and ranges from 0.20 to 0.50. The x-axis is labeled "Gini coefficient" and ranges from 25 to 45. Data points are labeled with country codes: SI, CZ, SI, IS, NO, DK, FI, SE, AT, DE, CH, AU, NZ, HR, JP, IT, GB-GBN, TH, IL, SR, US, CL, BG, RU.

(b) Gini and expected mobility

The y-axis is labeled "Expected upward mobility (share)" and ranges from 0.30 to 0.65. The x-axis is labeled "Gini coefficient" and ranges from 25 to 45. Data points are labeled with country codes: SI, CZ, IS, DK, NO, HR, SE, AT, DE, CH, AU, JP, FR, IT, RU, GB-GBN, TH, SR, IL, US, BG, CL.

1

Our proposed mechanism builds on models of anticipatory utility, in which individuals derive psychological benefit from believing in favorable future outcomes and weigh this benefit against the cost of poorer decisions based on distorted beliefs (Brunnermeier and Parker, 2005). Consider two societies with the same total income but different distributions: in the more unequal society, income is more concentrated at the top, so that the reward associated with reaching the highest position is larger. A larger top reward means that individuals have more to gain psychologically from believing they will obtain it, strengthening the incentive to distort beliefs upward. Holding the cost of belief distortion fixed, this yields a sharp prediction: higher inequality should induce greater overconfidence about one’s chances of reaching the top.

Testing this prediction in the same ISSP data, Panel (b) of Figure 1 shows that the share of respondents who expect to move up in 10 years is significantly *higher* in more unequal countries. Combined with the negative relationship between inequality and realized actual upward mobility, this produces a belief reversal: citizens in more unequal societies are more optimistic about moving up despite experienced less of it.¹

This belief reversal is consistent with our prediction, but it could reflect unobserved institutional or cultural differences across countries. To control for such differences, we exploit within-country variation in inequality exposure across birth cohorts. Following Roth and Wohlfart (2018), we measure inequality exposure as the top income concentration each respondent experienced during their formative years (ages 18 to 25) and estimate its effect on expected mobility controlling for country fixed effects, subregion-by-cohort fixed effects, family-of-origin social status, and six macroeconomic experiences (such as inflation and unemployment rates) during the same formative years that could independently shape expectations. A one standard deviation increase in formative-years experienced inequality is associated with an increase of 0.17 standard deviations in expected upward mobility ($p < 0.01$). Moreover, as our anticipatory utility model predicts, the increase in optimism concentrates at the top of the status distribution: by mapping ladder rungs to household income within each country, we find that experienced inequality raises the probability of expecting a destination that pays above mean income but lowers it for destinations that pay at or below the mean, even those that represent above-median status.

If mobility beliefs are shaped by inequality, they should in turn affect attitudes toward redistribution, as the POUM hypothesis predicts. We find that respondents who expect higher future status express weaker demand for redistribution and greater acceptance of inequality, controlling for current status and demographics. This pattern is consistent with a channel through which inequality raises optimism, which in turn lowers the political demand for correcting it.

¹This reversal is robust across multiple dimensions. It holds when inequality is measured by the top-5% income share instead of the Gini coefficient; when mobility is measured as a continuous gap between expected and current status, as status persistence as in Corak (2013), or as the share of respondents who expect to move from the bottom to the top of the social ladder; and when outlier countries such as South Africa and the Philippines are excluded from the estimation (Appendix B). The correlations also survive country-level controls such as average income and GDP growth.

In Section 4, we complement the survey evidence with an online experiment designed to test, in a controlled setting, whether inequality causally produces overoptimism through the motivated beliefs mechanism. Participants are randomly assigned to societies with either high or low initial inequality. Each society consists of five individuals: a single “Rich” individual and four “Poor” individuals. In the High inequality condition, the Rich earns \$10 while each Poor earns \$0; in the Low inequality condition, the Rich earns \$2.4 while each Poor earns \$1.9. After being informed of the initial inequality, all participants complete a quiz, with their ranking determined by a combination of quiz performance and luck. The highest-ranked individual, Rank 1, becomes the Rich. After taking the quiz but before learning their rank, participants report their beliefs about becoming Rich (with incentives for accuracy), along with their beliefs about obtaining each of the remaining ranks. This design allows us to test directly whether exogenous exposure to high inequality induces greater overoptimism about reaching the top.

We find that higher inequality causally increases overoptimism about reaching the top. The experiment allows us to cleanly define and measure overoptimism, because the objective probability of becoming Rich is known to be 20% for the average participant in both conditions. Subjects in both conditions display substantial overoptimism about Rank 1, believing they have over a 30% chance of becoming Rich. Higher inequality causes significantly more overoptimism: subjects in the high inequality condition report a 41.3% chance of achieving Rank 1, compared to 34.4% in the low inequality condition ($p < 0.01$), a 7 percentage point difference despite identical objective probabilities. This effect extends to the extensive margin: the share of subjects who believe they have an above-average chance of obtaining Rank 1 increases from 54.3% in the low inequality condition to 64.7% in the high inequality condition. This inequality-induced overoptimism cannot be explained by differences in effort or performance: we find no significant differences between conditions in quiz performance, time spent on the task, or self-reported effort levels (both own effort and beliefs about other’s effort).

Importantly, we find supporting evidence that the higher overconfidence observed in the high inequality condition is motivated rather than reflecting general optimism about performance. To test this mechanism, we examine how beliefs change across all five possible ranks. This provides a sharp test: if subjects in the high inequality condition simply believed they performed better on the quiz, this performance optimism would systematically affect beliefs about all above-median ranks. Since both Rank 1 and Rank 2 represent above-median outcomes in our five-person groups, believing one performed better should increase the probability assigned to both ranks. Using simulations based on our empirical performance distribution, we find that plausible improvements in believed performance would increase both mean and median Rank 2 beliefs substantially. We further show mathematically that at least half of subjects should see their Rank 2 beliefs increase under this alternative mechanism. Yet our data show the opposite pattern: while Rank 1 beliefs rise significantly in the high inequality condition at both the mean (41.3% vs. 34.4%) and the median (35.0% vs. 24.0%), Rank 2 beliefs remain unchanged at the median (20.0% vs. 20.0%) and actually decrease at the mean (18.6% vs. 20.6%). This selective increase, appearing only for the high-paying position, is inconsistent with performance optimism but aligns precisely with motivated beliefs about obtaining the high-paying position.

We further test the robustness of inequality-induced overconfidence by designing a Feedback treatment in which subjects receive accurate information about their quiz performance and statistical probability of achieving Rank 1 before stating their beliefs. If the observed overconfidence were driven by a lack of information rather than motivated reasoning, providing accurate feedback should eliminate the treatment difference in beliefs. We find that it does not: subjects in the high inequality condition still report higher Rank 1 beliefs than those in the low inequality condition (33.7% versus 29.3%, $p < 0.10$), even after receiving precise performance information. Because subjects in this treatment are told their objective probability, we can directly measure belief distortion as the deviation between reported and objective beliefs. The majority of subjects report beliefs higher than their objective probability, and this share is larger under high inequality, significantly so among low-performing subjects. While overall overconfidence is reduced relative to the Baseline treatment, the persistence of the inequality-driven gap in beliefs is consistent with motivated belief distortion: subjects maintain optimistic beliefs about reaching the top despite being confronted with accurate information about their relative performance.

The inequality-driven overoptimism causally raises inequality acceptance. In a second stage, before learning their actual rank, participants choose their preferred income allocation between the Rich and the Poor, with one randomly selected participant's choice implemented for the entire society. Subjects in the high inequality condition allocate on average 48.0% to the Rich, compared to 40.8% in the low inequality condition ($p < 0.01$). To establish that this effect runs through beliefs rather than through a direct shift in fairness views or anchoring, we exploit the Feedback treatment as a source of exogenous variation in beliefs. The feedback tells each subject their own objective probability of becoming Rich, so it moves beliefs in different directions depending on actual performance: it lowers the overoptimism of low performers and raises the confidence of high performers. Crucially, the share allocated to the Rich moves in the same direction wherever the feedback moves beliefs: down for low performers whose optimism is corrected, up for high performers whose confidence is confirmed. A formal instrumental variables analysis, using the interaction of feedback and quiz performance as the instrument, confirms that beliefs causally increase inequality acceptance. Together with the ISSP evidence that higher expected upward mobility is associated with lower demand for redistribution and greater acceptance of inequality, these results identify a channel through which inequality sustains itself: it induces overoptimism, which reduces the political demand for correcting it.

This paper contributes to the Prospect of Upward Mobility (POUM) literature (Piketty, 1995; Benabou and Ok, 2001) by endogenizing mobility beliefs. Previous work documents that mobility beliefs matter for redistribution preferences but does not explain why beliefs are systematically too optimistic in high-inequality societies. Alesina et al. (2018) show that Americans are more optimistic than Europeans about social mobility and that these beliefs correlate with lower redistribution demand, but do not address why citizens in more unequal societies are more optimistic despite facing lower actual mobility, as documented by the Great Gatsby Curve (Corak, 2013; Chetty et al., 2017). In Piketty (1995), agents rationally update beliefs based on their own mobility experience, so beliefs should on average track actual mobility, yet they do not. We resolve this tension: inequality itself shapes beliefs through a motivated channel. Because higher inequality increases the reward at the top, it

raises the anticipatory utility from believing one will get there (Brunnermeier and Parker, 2005), strengthening the incentive to distort beliefs upward. Our mechanism thus provides a microfoundation for the POUM hypothesis that is consistent with the Great Gatsby Curve, and offers an explanation for the persistent overconfidence in upward mobility in the US (Davidai and Gilovich, 2015).

A natural question is whether information can correct these belief distortions. Our findings speak to the growing literature on information provision and belief updating about economic conditions, in which a series of influential experiments provide subjects with accurate information about inequality, income rank, or tax policy, studying the effects on beliefs and preferences (Cruces et al., 2013; Kuziemko et al., 2015; Karadja et al., 2017; Stantcheva, 2021; Hvidberg et al., 2023). This paradigm treats misperceptions as correctable: providing information should update beliefs. Our framework offers a different perspective. Because belief distortions serve a psychological function, the anticipatory utility from expecting to reach the top, they may resist correction. Our feedback treatment provides direct evidence for this distinction: subjects given their individual-specific objective probability of reaching Rank 1 still display significantly higher overconfidence in the high-inequality condition, ruling out not only information absence but also information frictions as the driver of the belief gap. This complements Kuziemko et al. (2015), who find that information about inequality moves perceptions but barely shifts policy preferences, and Hoy and Mager (2021), who document the striking result that individuals told they are poorer than expected become *less*, not more, concerned about inequality. Together, these patterns suggest that belief distortions about economic prospects can operate through motivated channels that information alone cannot address. Moreover, using US data from the General Social Survey, we show that optimistic expectations about future social status are positively associated with current happiness, controlling for actual status, especially among low-income individuals. This suggests that the very beliefs that sustain inequality also provide psychological benefits to those disadvantaged by it, raising a welfare dilemma: correcting overoptimistic mobility beliefs may improve democratic decision-making but reduce the wellbeing of those who need it most.

We also contribute to the literature on motivated beliefs and overconfidence. A large body of research documents that individuals hold overconfident beliefs driven by motivational factors (Bénabou and Tirole, 2002; Brunnermeier and Parker, 2005; Eil and Rao, 2011; Schwardmann and van der Weele, 2019; Zimmermann, 2020; Huffman et al., 2022). This literature has primarily focused on how *intrapersonal* factors, such as ego concerns, risk preferences, and prior performance, shape motivated beliefs. We shift the focus to a *macroeconomic* determinant: we show that the degree of motivated overconfidence responds to the level of inequality in the economic environment. Furthermore, while anticipatory utility is central to theoretical accounts of why individuals distort beliefs (Brunnermeier and Parker, 2005), direct evidence that distorted beliefs actually deliver felicity has been scarce; our happiness result provides field evidence for this channel.

Our individual-level survey evidence contributes to the growing literature on how macroeconomic experiences during formative years persistently shape economic beliefs and behavior. Malmendier and Nagel (2011) establish the foundational result that individuals who ex-

perienced low stock market returns during their youth take less financial risk throughout their lives. Subsequent work has extended this to inflation expectations (Malmendier and Nagel, 2016), consumption behavior (Malmendier and Shen, 2024), and political preferences (Alesina and Fuchs-Schündeln, 2007); Giuliano and Spilimbergo (2025) provide a comprehensive survey. Most closely related to our work, Roth and Wohlfart (2018) demonstrate that experienced macroeconomic conditions, including unemployment and inequality, during one’s impressionable years persistently affect redistribution preferences. We extend this approach by showing that experienced inequality shapes not just redistribution preferences but *mobility beliefs*: individuals who experienced higher top income shares during ages 18–25 hold more optimistic expectations about reaching the top of the social ladder, even decades later. Moreover, this optimism is concentrated at the top of the distribution, consistent with our motivated beliefs mechanism rather than a uniform shift in expectations.

Our paper also contributes to the literature on social preferences and the demand for redistribution.² This literature has established that the inequality people are willing to accept depends on its source and on the fairness views they hold. We add a determinant that operates upstream of these fairness judgments: the level of inequality itself. Because higher inequality raises the anticipatory value of reaching the top, it raises optimism about getting there, and in our experiment this induced optimism causally raises the inequality people accept. In the cross-country survey, higher expected upward mobility is in turn associated with lower demand for redistribution. Inequality acceptance thus responds not only to how inequality arises but also to a motivated belief about where one expects to end up, a channel through which the economic environment itself shapes social preferences.

Finally, our cross-country survey evidence contributes to the literature on the Great Gatsby Curve. As coined by Krueger (2012), this literature documented that children in more unequal societies face lower chances of moving up the income ladder (Corak, 2013; Chetty et al., 2017; Durlauf and Seshadri, 2018; Durlauf et al., 2022). We complement this by documenting a “belief reversal” of the Great Gatsby Curve: while actual mobility is lower in more unequal societies, *believed* future mobility is higher. Citizens in more unequal countries are more likely to expect upward movement on the social ladder, and this optimism is concentrated at the very top of the status distribution. This divergence between actual and believed mobility has implications for how we interpret the policy consequences of the Great Gatsby Curve: if citizens in high-inequality societies are more optimistic about their prospects despite facing lower actual mobility, then inequality may be more politically sustainable than the objective mobility data alone would suggest.

The remainder of the paper proceeds as follows. Section 2 presents our conceptual framework. Section 3 presents the cross-country survey evidence from the International Social

²A large literature has studied the nature of social preferences (Fehr et al., 1993; Rabin, 1993; Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Andreoni and Miller, 2002; Charness and Rabin, 2002; Engelmann and Strobel, 2004; Bellemare et al., 2008; Bartling et al., 2015; Cappelen and Tungodden, 2019; Fehr and Vollmann, 2022; Exley and Kessler, 2024), including how the source of inequality shapes inequality acceptance (Konow, 1996, 2000; Alesina et al., 2001; Cappelen et al., 2007; Konow, 2009; Cappelen et al., 2010; Krawczyk, 2010; Balafoutas et al., 2013; Cappelen et al., 2013; Bauer et al., 2014; Durante et al., 2014; Mollerstrom et al., 2015; Akbaş et al., 2019; Cassar and Klein, 2019; Almås et al., 2020; Sugden and Wang, 2020; Müller and Renes, 2021; Cappelen et al., 2022; Barr et al., 2023; Fehr et al., 2024).

Survey Programme. Section 4 describes our experimental design and Section 5 our main experimental results. Section 6 concludes.

2 Conceptual framework

We develop a simple model of motivated belief formation based on the optimal expectations framework of Brunnermeier and Parker (2005). In this framework, forward-looking individuals derive well-being not only from realized outcomes but also from anticipation of future outcomes. Since optimistic beliefs enhance current well-being, individuals have an incentive to distort their beliefs upward. However, distorted beliefs lead to worse decisions, and thus lower realized outcomes, creating a cost of distortion. Optimal beliefs balance these two forces. A key insight of Brunnermeier and Parker (2005) is that, starting from rational beliefs, a small bias leads to *first-order* gains in anticipatory utility but only *second-order* costs from distorted decisions, so it is always optimal to be at least slightly optimistic. We show that this distortion responds to the economic environment: higher inequality amplifies the incentive to hold optimistic beliefs about reaching the top.

Setup. Consider a society with n individuals. There are n ranked positions $j \in \{1, 2, \dots, n\}$, where the top k positions (“Rich”) each pay income y_R and the bottom $n - k$ positions (“Poor”) each pay $y_P < y_R$. Total income is $Y = k \cdot y_R + (n - k) \cdot y_P$, and mean income is $\bar{y} = Y/n$. Income inequality is captured by π , the share of total income going to Rich positions, so that $y_R = \pi Y/k$ and $y_P = (1 - \pi)Y/(n - k)$. Higher π implies greater inequality.

Each individual i has an objective probability p_{0ij} of ending up in position j , which may vary across individuals due to differences in ability. Before learning their position, individuals form beliefs about the probability of occupying each position.

Optimal beliefs. Following Brunnermeier and Parker (2005), each individual chooses subjective beliefs $\boldsymbol{\mu}_i = (\mu_{i1}, \dots, \mu_{in})$ over positions to maximize well-being, which trades off anticipatory utility against the cost of distorted decisions:³

$$W_i(\boldsymbol{\mu}_i) = \underbrace{\sum_{j=1}^n \mu_{ij} \cdot u(y_j)}_{\text{anticipatory utility}} - \underbrace{\sum_{j=1}^n \alpha (\mu_{ij} - p_{0ij})^2}_{\text{decision cost}} \quad \text{s.t.} \quad \sum_{j=1}^n \mu_{ij} = 1 \quad (1)$$

The first term is anticipatory utility: the felicity individuals experience from contemplating their future income under subjective beliefs. Holding a higher belief about occupying a high-income position increases current well-being. The second term captures the cost of belief distortion. As in Brunnermeier and Parker (2005), distorted beliefs lead to suboptimal decisions, reducing realized outcomes. The quadratic form with $\alpha > 0$ is a tractable speci-

³The optimization over $\boldsymbol{\mu}_i$ is an “as if” characterization of the outcome of a belief formation process. As Brunnermeier and Parker (2005) noted, their “optimal expectations framework is a simple model that captures complex brain processes” regardless of whether the mapping from objective to subjective probabilities is deliberate or automatic.

fication; our qualitative results hold under any strictly convex cost that is minimized at the objective probability.⁴

Assuming linear utility $u(x) = x$ for tractability and an interior solution, the Lagrangian for the constrained optimization yields the first-order condition for each position j :

$$y_j - 2\alpha(\mu_{ij} - p_{0ij}) = \lambda \quad (2)$$

where λ is the multiplier on the adding-up constraint. Summing over j and using $\sum_j \mu_{ij} = \sum_j p_{0ij} = 1$ gives $\lambda = \bar{y}$, the mean income across positions. The optimal belief distortion for position j is therefore:

$$\mu_{ij}^* - p_{0ij} = \frac{y_j - \bar{y}}{2\alpha} \quad (3)$$

This expression has a clean interpretation: the belief distortion for each position is proportional to how much that position's income deviates from the mean. Individuals become overconfident about positions that pay above-mean income ($y_j > \bar{y}$), and underconfident about positions that pay below-mean income ($y_j < \bar{y}$).

Note the key asymmetry from Brunnermeier and Parker (2005): at the objective beliefs $\mu_{ij} = p_{0ij}$, the marginal benefit of shifting probability toward a high-income position is $y_j - \bar{y}$ (first order), while the marginal cost $2\alpha(\mu_{ij} - p_{0ij}) = 0$ (second order). It therefore always pays to distort beliefs at least slightly toward optimism about high-income positions.

How inequality affects beliefs. Consider an increase in inequality π holding total income Y fixed, so that $y_R = \pi Y/k$ increases, $y_P = (1 - \pi)Y/(n - k)$ decreases, and $\bar{y} = Y/n$ remains unchanged. From Equation 3, for Rich positions ($j \leq k$), the distortion $\frac{y_R - \bar{y}}{2\alpha}$ is increasing in π , since $\frac{\partial y_R}{\partial \pi} = Y/k > 0$. While for Poor positions ($j > k$), the distortion $\frac{y_P - \bar{y}}{2\alpha}$ is decreasing in π , since $\frac{\partial y_P}{\partial \pi} = -Y/(n - k) < 0$.

This yields two testable predictions:

Prediction 1: Higher inequality increases overconfidence about reaching high-income positions.

Prediction 2: Higher inequality does not increase beliefs about positions that pay below the mean, even if those positions are above the median rank.

Prediction 2 provides a sharp test to distinguish motivated beliefs from general performance optimism. Under performance optimism, individuals in high-inequality environments believe they performed better on a task determining rank. The general intuition of this alternative is that higher believed performance should lead to a more across-the-board increase in confidence about above-median positions.⁵ The optimal expectations model, by contrast,

⁴Alternative specifications include the Kullback-Leibler divergence used in Caplin and Leahy (2019). In our experiment, the Binarized Scoring Rule used to incentivize belief reports provides a concrete source of decision cost: inaccurate reports reduce the expected accuracy payment, with the expected loss taking a quadratic form.

⁵The exact prediction of performance optimism for each individual rank depends on the performance distribution and the magnitude of believed improvement. We provide a detailed analysis in Appendix E.1.

predicts that belief distortion targets specifically those positions with above-mean income. In our setting with $k < n/2$ (fewer Rich than Poor), some above-median-rank positions still pay below-mean income. These positions allow us to test the two theories apart: the model predicts that beliefs about positions that rank above the median but pay below the mean income should *not* increase with inequality, and may in fact decrease, while performance optimism generally predicts an increase.

Robustness of the framework. The baseline model assumes that subjects form beliefs without anticipating the downstream allocation decision, which matches our experimental Baseline, where Period 2 is introduced only after beliefs are stated. In Appendix F we show that both predictions survive when subjects fully anticipate the Period-2 allocation, with beliefs determined in a fixed point together with the allocation they expect to implement. Subjects remain strictly overly optimistic about reaching the top, and higher Period-1 inequality continues to raise both the equilibrium belief and, through it, the inequality that subjects implement in Period 2. The mechanism is therefore not an artifact of subjects’ ignorance of the downstream decision; it operates even under full awareness.

3 Evidence from ISSP Survey data

Our conceptual framework predicts a positive relationship between inequality and beliefs about reaching the top of the income distribution. We first examine this prediction in large-scale cross-country survey data, which captures the complex dynamics of real-world environments in which individuals have long-term exposure to inequality and face actual economic outcomes. We then ask whether the same pattern holds within countries, exploiting variation in the inequality that different birth cohorts experienced during their formative years.

We use data from the International Social Survey Programme (ISSP).⁶ The ISSP 2019 “Social Inequality V” module asks respondents to place themselves on a social ladder running from 1 (bottom) to 10 (top) at three points in time: where their family stood when they were growing up (v_{42}), where they stand now (v_{41}), and where they expect to be in ten years (v_{43}).⁷ These three questions let us construct, for each respondent, a measure of *actual* upward mobility (current status minus past status, $v_{41} - v_{42}$) and a measure of *expected* upward mobility (expected status minus current status, $v_{43} - v_{41}$). This is what makes the module uniquely suited to our question: within the same population, it lets us compare how inequality relates to mobility that has already happened and to mobility that people anticipate.

⁶Data were collected between November 2018 and May 2022, depending on the country. The ISSP provides individual weights to improve representativeness. The results in this section do not use these weights but are qualitatively and quantitatively robust to including them (available on request). Summary statistics split by countries with high and low Gini coefficient are in Appendix Table C.1.

⁷The exact wording is, for current status, “In our society there are groups which tend to be towards the top and groups which tend to be towards the bottom. Below is a scale which runs from top to bottom. Where would you put yourself on this scale?”; for past status, “Where did the family that you grew up in fit in then?”; and for expected status, “Thinking ahead 10 years from now, where do you think you will be on this scale?”

3.1 The Great Gatsby Curve and its belief reversal

Our framework predicts that in more unequal countries, where the gap between the top of the distribution and the average is larger, people hold more optimistic beliefs about moving up. Panel (b) of Figure 1 in the introduction bears this out. Across the 27 countries in our sample, average expected upward mobility is higher where inequality is higher; the cross-country correlation between the Gini coefficient and expected mobility is positive, 0.60 ($p < 0.01$).

This relationship is the mirror image of the Great Gatsby Curve. Panel (a) of Figure 1 is a conceptual replication of that curve (Corak, 2013): in more unequal countries, the average respondent reports less actual upward movement from family status to own status, a cross-country correlation of -0.50 ($p < 0.01$). The two relationships carry opposite signs even though both describe mobility on the identical 1 to 10 ladder for the same respondents. Inequality is associated with less mobility in the rear-view mirror and more mobility looking forward. Beliefs thus supply, through expectations, exactly the positive inequality and mobility link that the prospect of upward mobility hypothesis requires but that the realized data deny (Benabou and Ok, 2001).

The forward-looking optimism is concentrated where our framework says it should be, at the top of the ladder, where the rewards from reaching it are largest. Comparing the full distribution of expected status between more equal countries (Gini below the median) and more unequal countries (Gini above the median), the two distributions look alike over the bottom and middle of the ladder and diverge only at the top. The predicted share expecting a top-two position rises from about 7% in more equal countries to about 11% in more unequal ones (Appendix Table C.4 and Figure C.8). This is a thickening of the upper tail of expectations rather than a uniform upward translation of the whole distribution, and it does not appear in how respondents place themselves *currently*: in unequal countries, respondents are if anything more likely to place themselves in the bottom half today (Appendix Figure C.9). The optimism is specific to the top of the future distribution.

The optimism also runs ahead of realized experience. At the country level, average expected mobility exceeds average actual mobility, a gap of 0.063 on the binary measure of moving up, and the gap is larger in more unequal countries: the correlation between the gap and the Gini coefficient is 0.61 on the binary measure ($p < 0.01$), and the continuous ladder-difference measure gives the same pattern. The most overoptimistic countries are the Philippines, Chile, and Russia; the least overoptimistic are France, Czechia, and Finland. The full ranking and the underlying correlations are reported in Appendix Table B.2.

Robustness. The belief reversal does not depend on how inequality is measured, how mobility is measured, or on a few influential countries; details are in Appendix B. First, it holds under alternative inequality measures: replacing the Gini coefficient with the top-5% or top-1% income share leaves both signs unchanged (Appendix Figure B.2). Second, it holds under alternative mobility measures, with actual mobility falling and expected mobility rising in inequality whether mobility is the continuous ladder difference, Corak status persistence, or a Chetty rags-to-riches transition rate (Appendix Figures B.3, B.4, and B.5). Third, it is not driven by the two highest-leverage countries: excluding South Africa and the Philippines

leaves the reversal intact (Appendix Figure B.1). Fourth, it survives country-level controls for average income, growth, and institutions, and with the full control set the cross-country relationship strengthens rather than weakens (Appendix Figure B.6 and Table B.1).

Result 1: *Across countries, higher inequality is associated with higher expected upward mobility but lower actual mobility, and this optimism is concentrated at the top of the ladder.*

3.2 Experienced inequality during formative years

The cross-country comparison is suggestive, but countries differ in many ways besides inequality, and a raw correlation across 27 societies cannot separate inequality from everything that travels with it. To bring the comparison inside countries, we exploit variation *within* countries, across birth cohorts, in the inequality that people were exposed to when they were young. Following Roth and Wohlfart (2018), we measure experienced inequality as the average top-5% income share in a respondent’s country over the ages 18 to 25, the “impressionable years” during which economic experiences leave a lasting imprint on beliefs (Malmendier and Nagel, 2011; Giuliano and Spilimbergo, 2025). Because different cohorts in the same country came of age under different levels of top income concentration, this measure varies within a country even after we absorb everything fixed about the country itself. We standardize it within the sample (one standard deviation is 9.5 percentage points of top-5% share, roughly the gap between Northern Europe and the United States) and restrict the sample to respondents aged 29 or older, so that the formative window has closed for everyone we study.⁸

Figure 2 reports how experienced inequality relates to expected mobility, $v_{43} - v_{41}$, across four specifications, with the underlying coefficients in Appendix Table C.2. We use expected mobility rather than the level of expected status because the level mechanically confounds optimism with current standing: a cohort that already places itself higher will report a higher expected status for that reason alone, and differencing removes this contamination. The estimating equation is

$$(v_{43,i} - v_{41,i}) = \beta \overline{\text{top5}}_{c(i),b(i)}^{\text{std}} + X_i' \gamma + \alpha_{c(i)} + \delta_{r(c(i)) \times b(i)} + \varepsilon_i,$$

where $\overline{\text{top5}}_{c(i),b(i)}^{\text{std}}$ is standardized formative-years experienced inequality for cohort $b(i)$ in country $c(i)$, one standard deviation of which is 9.5 percentage points of top-5% share. The vector X_i collects three individual controls: age, an indicator for female, and years of education. Birth cohorts differ systematically in their age structure, their schooling, and their gender composition, and each of these independently moves mobility expectations, so absorbing them ensures the coefficient on experienced inequality is not simply picking up cohort composition. The term $\alpha_{c(i)}$ is a country fixed effect, which absorbs everything fixed about a country: its culture, its institutions, its average level of inequality, and its average optimism. The term $\delta_{r(c(i)) \times b(i)}$ is a subregion-by-cohort fixed effect, which lets the age and cohort profile of expectations differ across world regions, so that identification comes only

⁸For respondents younger than 29, the 18 to 25 window overlaps with the present, so the measure no longer captures exposure that is firmly in the past. Restricting to age 29 and older ensures the formative experience precedes the survey for the entire sample.

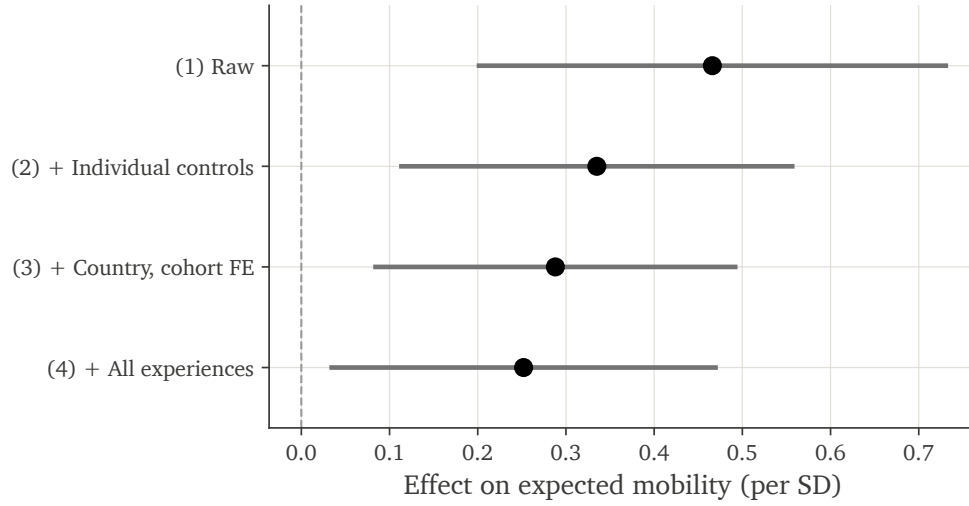
from within-country, across-cohort variation in experienced inequality net of any region-wide cohort trend. Standard errors are clustered at the country level.

The most demanding specification adds a further set of controls, each a country-level average over the same ages 18 to 25 window: the unemployment rate, GDP-per-capita growth, CPI inflation, the level of electoral democracy, exposure to a banking or currency crisis, and exposure to a rare disaster, together with the respondent’s family-of-origin status (reported parental status, v_{42}). A large body of work shows that macroeconomic experiences during the impressionable years leave a lasting imprint on economic beliefs and risk-taking (Malmendier and Nagel, 2011; Giuliano and Spilimbergo, 2025): inflation experienced when young shapes later inflation beliefs (Malmendier and Nagel, 2016), recessions and aggregate shocks shape preferences and beliefs about luck and redistribution (Giuliano and Spilimbergo, 2025), adverse financial shocks leave durable scars (Malmendier and Shen, 2024), and coming of age under a different political regime shapes redistribution preferences (Alesina and Fuchs-Schündeln, 2007). Any of these experiences could have co-moved with top income concentration when a cohort was young, so we hold them fixed and ask whether experienced inequality still predicts mobility expectations on its own.

The coefficient is positive and barely moves as we add successively more demanding controls. Specification (1) of Figure 2 regresses expected mobility on standardized experienced inequality alone, with no controls: a one standard deviation increase in formative-years experienced inequality is associated with 0.466 ladder rungs higher expected mobility (SE 0.135, $p < 0.01$, $N = 22,558$). Specification (2) adds the three individual controls, age, an indicator for female, and years of education, so that the estimate no longer reflects differences in age structure, schooling, or gender composition across cohorts; the coefficient falls modestly to 0.335 (SE 0.113, $p < 0.01$, $N = 21,879$). Specification (3) further adds country and subregion-by-cohort fixed effects, which is the identifying comparison of interest: it contrasts adjacent cohorts within the same country, net of any region-wide cohort trend, rather than comparing across countries that differ in many other ways. The coefficient is 0.288 (SE 0.104, $p < 0.01$, $N = 21,879$), which translates to about 0.17 standard deviations of expected mobility. Column (4) adds the six macroeconomic experiences measured over the same ages 18 to 25 window, the unemployment rate, GDP-per-capita growth, CPI inflation, electoral democracy, banking-or-currency-crisis exposure, and rare-disaster exposure, together with family-of-origin status, so that experienced inequality cannot be standing in for some other formative-years shock that happened to co-move with it; the coefficient is 0.252 (SE 0.111, $p < 0.05$, $N = 19,152$) and remains significant. Across all four columns the estimate is positive, stable, and statistically significant at the 5% level. Individuals who came of age under higher inequality are associated with more expected upward mobility decades later, even compared with people from the same country from adjacent cohorts.

The framework makes a sharper prediction than “more optimism on average.” Belief distortion should be positive for destinations that pay above the population mean and negative for those that pay below it, even when the below-mean destinations rank above the median. We test this in income terms by mapping each ladder rung to the average household income of respondents at that rung within their country (Appendix Figure C.3). The mapping places the country-mean income between rungs 7 and 8, so that expecting rung 8 or higher means

Figure 2: Experienced inequality and expected mobility across specifications



Notes: Coefficient on standardized experienced inequality (formative-years top-5% income share, the average top-5% income share in the respondent's country over ages 18 to 25; one standard deviation = 9.5 percentage points) with 95% confidence intervals clustered at the country level. The outcome is expected mobility, expected status in ten years minus current status ($v_{43} - v_{41}$), on the 1 to 10 social ladder. Each specification adds to the previous one: (2) individual controls; (3) country and subregion-by-cohort fixed effects; (4) six lifetime macroeconomic experiences and family-of-origin status. The underlying regression estimates are reported in Appendix Table C.2. Sample: ISSP 2019 respondents aged 29 or older.

expecting a destination that pays above the country mean. Experienced inequality is associated with a 6.0 percentage point higher probability of expecting a destination that pays above the country mean, with an exactly offsetting drop in the probability of expecting a destination at or below it ($p < 0.01$; Appendix Figure C.4). A continuous version, which relates the income gap of the expected destination to experienced inequality, gives the same answer in a single number: a one standard deviation increase in experienced inequality is associated with the expected destination's income being 5.5% higher relative to the country mean ($p < 0.01$). The probability mass associated with higher inequality is not spread evenly across the ladder; it concentrates on the destinations that pay above the mean, consistent with motivated belief distortion and in contrast to a uniform rise in optimism.

Robustness. We conduct four robustness checks; details are in Appendix C.1. First, the pattern might be driven by a single generation whose formative years happened to coincide with a particular episode. It is not: estimating the relationship separately for seven age cohorts (Appendix Figure C.5) yields a positive coefficient in every cohort, significant across the younger and middle-aged groups and present, if noisier, among the oldest. Second, the result might depend on the specific top-5% measure: it does not, as the top-1% and top-10% income shares give the same pattern (Appendix Figure C.6). Third, the relationship is linear, with no kink or threshold (Appendix Figure C.1). Fourth, and most directly, a placebo that uses inequality experienced *before* the formative window (ages 0 to 7) produces a precisely estimated zero, as does a post-formative window (ages 34 to 41); only the 18 to

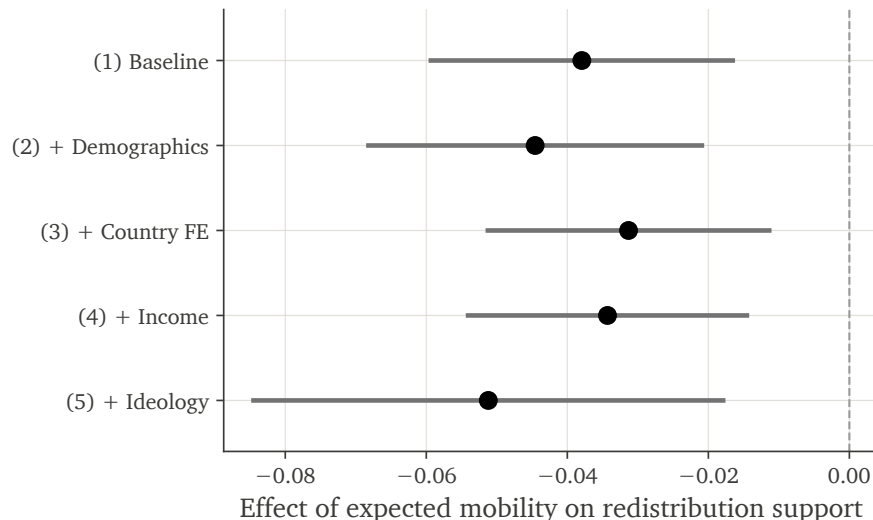
25 window carries a positive, significant coefficient (Appendix Figure C.7). Every coefficient in this section reproduces across Python and Stata.

Result 2: *Within countries, individuals who experienced higher inequality during their formative years expect more upward mobility decades later, and this optimism is concentrated on destinations that pay above the mean income.*

3.3 Consequences for demand for redistribution

For optimism to sustain inequality, it must bear on what people want from policy. At the individual level, holding current status fixed, people who expect to move further up the ladder support less redistribution, the link the prospect of upward mobility hypothesis requires (Benabou and Ok, 2001). Figure 3 plots the coefficient on expected mobility across specifications that successively add demographics, country fixed effects, household income, and political ideology; the relationship stays negative and significant throughout ($p < 0.01$). The pattern parallels the experiment, where greater confidence about becoming Rich predicts allocating a larger share to the Rich: respondents who expect to rise are less willing to support policies that redistribute away from the top they expect to join.⁹

Figure 3: Expected mobility and demand for redistribution



Notes: Coefficient on individual expected upward mobility (expected status in ten years minus current status, $v_{43} - v_{41}$) from a regression of redistribution support on expected mobility, holding current status fixed. Redistribution support is agreement that government should reduce income differences between the rich and the poor, reverse-coded so higher values indicate stronger support. Each specification adds to the previous one: (2) demographics (age, gender, years of education); (3) country fixed effects; (4) household-income quintile; (5) left-right political ideology. 95% confidence intervals from standard errors clustered at the country level. Sample: ISSP 2019 respondents.

⁹Demand for redistribution is measured by agreement that government should reduce income differences between the rich and the poor, rescaled so that higher values indicate stronger support for redistribution.

Result 3: *Higher expected mobility is associated with weaker demand for redistribution, the link the prospect of upward mobility hypothesis requires.*

Taken together, the survey evidence traces the full chain our framework describes operating in the field: higher inequality correlates with more optimistic mobility beliefs, that optimism concentrates at the top, and more optimistic beliefs correlate with weaker demand for redistribution. To test the mechanism causally, we next turn to an experiment that manipulates inequality directly and measures beliefs at the individual level, in a setting where inequality, beliefs, and institutions are not jointly determined as they are across societies.

4 Experimental design

This study employs a between-subjects experimental design.¹⁰ We first present the Baseline treatment, which consists of two conditions: high inequality and low inequality. We then introduce the Feedback treatment, which differs from the Baseline by providing subjects with performance feedback after completing the quiz.

Overview. The experiment consists of two periods. In Period 1, subjects face a fixed income allocation that creates inequality between one “Rich” and four “Poor” members in their five-person group. Who becomes rich or poor depends on their relative ranking, determined by performance on a quiz with some random noise. Under this given inequality, subjects work on the quiz. After they finish the quiz but before they know their rankings, we elicit their beliefs about their chances of becoming rich. In Period 2, subjects choose their preferred level of inequality between the Rich and the Poor. They choose their preferred allocation knowing that the ranking from Period 1 will also determine their position in Period 2, and still without knowing their actual ranking. Thus, subjects make their inequality choice under beliefs formed in Period 1. Only after making this choice, we determine subjects’ rankings and payoffs in both periods. One period is randomly selected for payment.

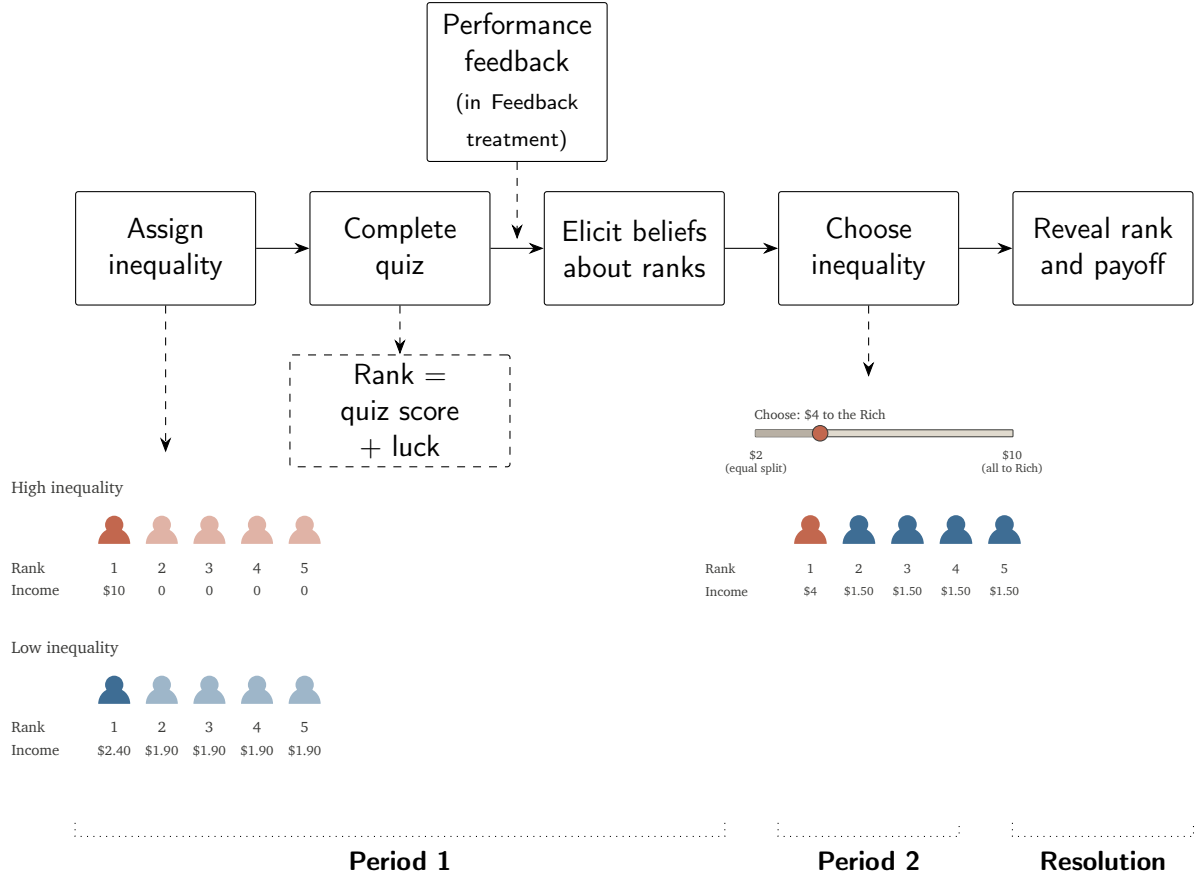
The main treatment manipulation is to assign either a high or low income inequality to a group in Period 1. Figure 4 provides an overview over the main experimental stages.¹¹

The experiment is deliberately abstract. Rather than reproduce the institutions of a real economy, with its labor markets, taxes, and life-cycle dynamics, we isolate the single psychological channel our framework concerns: how the level of inequality a person faces shapes the belief they form about reaching the top. Two design choices follow. First, the experimental society contains no mobility process to learn from and no repeated play. Real economies give people decades of feedback about their own and others’ mobility, and it is precisely this accumulated experience that makes field beliefs hard to interpret. We abstract it away so that the belief a subject reports is formed under a known inequality, not contaminated by an idiosyncratic mobility history. The aim is to identify the mechanism cleanly, not to match

¹⁰The design was approved by the institutional review board of WZB on July 26 2023 (Ethics Application No. 208) and pre-registered at the AEA RCT Registry under AEARCTR-0014848.

¹¹In the experimental implementation, there are two variations of the Baseline treatment, in which we varied when exactly subjects completed the main part of the quiz. As pre-registered, we pool these two variations since there were no significant differences between them. Details are described in Appendix D.

Figure 4: Experimental design and timeline



Notes: The panels below *Assign inequality* show the Period-1 income at each of the five ranks under the High and Low inequality conditions. The panel below *Choose inequality* illustrates the Period-2 allocation: subjects choose how much of the \$10 goes to the Rich (here \$4), with the remainder split equally among the four Poor (here \$1.50 each). Dashed elements mark the rank, which is determined by quiz performance and luck once the quiz ends but stays hidden until the end: subjects state beliefs about each rank without knowing it, the same rank carries into Period 2, and it is revealed only at payment. The belief about being Rank 1, the Rich, is incentivized. One of the two periods is paid at random. Subjects are aware of the existence of period 2 in period 1, but they do not know about the choice they need to make in period 1. The feedback provision occurs only in the Feedback treatment.

the texture of the field setting that motivates it. This is also why we do not build mobility into the design: introducing it would reintroduce exactly the confound the experiment exists to remove.

Second, the clean setting lets us measure overoptimism precisely, which the survey cannot.¹² We set the objective probability of becoming Rich and elicit the belief, so overoptimism is simply the gap between the two. We measure this gap at the group level in the Baseline treatment, where the average objective probability is 20%, and at the individual level in the Feedback treatment, where each subject is told their own objective probability.

Period 1 income inequality. In the experimental “society”, there are always one Rich subject (referred to as the “high income earner” in the instructions) and four Poor subjects (referred to as the “low income earner” in the instructions). Who becomes the Rich is determined by their ranking in the society. Subjects’ incomes by rank in each inequality condition are shown in the income panels of Figure 4.

In the “High” inequality condition, the Rich (Rank 1) earns \$10, while the Poor (Ranks 2-5) earn \$0. In the “Low” inequality condition, the Rich earns \$2.4, while the Poor (Ranks 2-5) earn \$1.9. Total group income is \$10 in both conditions, so mean income is \$2 in both: the manipulation changes only the concentration of income at the top, not its level. This matches the constant-total-income comparative static in our framework and ensures the treatment varies inequality rather than the scale of stakes. The income structure in Period 1 is announced at the beginning of the experiment. Subjects are not aware of the other inequality condition.

Ranking. A subject’s ranking (Rank 1 vs. Ranks 2-5) is determined by quiz performance or luck. With 80% probability, rankings are assigned based on performance in a quiz. The quiz consists of solving 10 Raven matrices within a 10-minute time limit. Performance is defined as the number of correct answers. Ties in performance are broken randomly in ranking. Subjects are informed about how their earnings depend on their relative ranking and there is no additional payment based on their absolute performance. Before working on the 10 quiz questions, we also provide two example Raven matrices to familiarize them with this type of task, but they do not count towards their performance.

With 20% probability, rankings are assigned randomly, giving each subject an equal chance of obtaining any ranking.

Belief elicitation. We elicit subjects’ beliefs about their ranking within their five-person group after the inequality announcement in Period 1 and completing the Raven matrix quiz questions.

Specifically, we ask subjects: “For each possible rank, what do you think is the probability that you will achieve it?” Subjects must assign a probability to each rank, with the probabilities summing to 100.

¹²In the survey, expected mobility is a belief about a future that has not yet happened, so it cannot be labeled overoptimistic on its own terms; the most one can do is benchmark it against realized past mobility. That benchmark is suggestive: in all 27 ISSP countries the average respondent expects to climb the ladder over the coming decade, with mean expected mobility of 0.44 rungs against 0.35 rungs of realized mobility, and on a relative ladder, where not everyone can move up, universally positive expectations are difficult to square with accuracy. The comparison is nonetheless imperfect, because the future need not mirror the past. The experiment removes the ambiguity.

Beliefs about being in Rank 1 are incentivized with the Binarized Scoring Rule (BSR) (Hosain and Okui, 2013). The BSR computes the distance between the subject’s stated probability for being in Rank 1 and whether they actually are in Rank 1; the bonus is paid if the distance is smaller than a randomly drawn cut-off threshold, which makes truthful reporting the best strategy regardless of risk preferences. In the instructions, we follow the protocol recommended by Danz et al. (2022), i.e. we tell subjects that stating their true belief would maximize their chances of getting the bonus payment, while putting technical details of the payment rule behind an optional button.

Period 2 income allocation choice. In Period 2, subjects choose how to allocate income within their five-person group. Specifically, they decide how much of the total \$10 to allocate to the Rich, with the remainder divided equally among the four Poor. The set of options lies between \$2, i.e. fully equalizing income, and \$10, i.e. giving everything to the Rich and nothing to the Poor. One randomly selected group member’s allocation choice is implemented for the entire group.

When making this choice, subjects do not know their Period 2 ranking but are informed that it will be identical to their Period 1 ranking. This implies that they should base their choice on their previously formed Rank-1 belief. We remind them of their previously stated Rank-1 belief on the allocation decision page.¹³

After Period 2, we collect survey data to understand the factors influencing participants’ choices. Participants indicate how much various factors influenced their distribution decisions, assess the relative importance of effort, luck, and intelligence in quiz performance.¹⁴ We also gather standard socio-demographic and economic variables (gender, age, education, income, nationality, ethnicity, political orientation), as well as general redistribution and risk preferences. Additionally, we elicit participants’ beliefs about their own economic mobility by asking about their current relative income status and their projected status in 10 years.

After the session is complete, we determine the ranking that applies to both periods, calculate Period 1 payoff based on rankings and Period 2 payoff based on rankings and the chosen allocation, and randomly select one period for payment.

Feedback treatment. The Feedback treatment serves two purposes. First, it tests whether the inequality-induced overconfidence reflects missing information rather than motivated reasoning: if subjects are simply uninformed, giving them their true chance of becoming Rich should erase the inequality gap in beliefs. Second, and central to identifying the belief channel, the feedback generates *exogenous* variation in beliefs that we can use to instrument them. The feedback is individualized, and this is what makes it useful as an instrument: it tells each subject their own objective probability, which is low for those who performed

¹³As we do not elicit beliefs a second time, we cannot rule out that participants adjust their Rank-1 belief consciously or unconsciously upon learning about the possibility to redistribute. However, the fact that their Rank-1 belief is now more consequential to their payoffs should rather mitigate motivated beliefs (Zimmermann, 2020), which would reduce the treatment effect on inequality.

¹⁴As a robustness check, we also ask whether subjects have used AI during the quiz and whether they think others might have used AI. Only a minority of subjects admitted to having used AI (1%) or thought others might have used AI during the quiz (7%), and our results remain qualitatively unchanged if we drop them.

poorly and high for those who performed well. It therefore moves beliefs in opposite directions depending on performance, lowering the optimism of subjects who overestimated themselves and confirming or raising the beliefs of subjects who did well. We exploit this performance-dependent shift by examining the treatment effect of feedback conditional on actual quiz performance, which forms the first stage of the instrumental-variables analysis in Section 5.5.

The Feedback treatment is identical to the Baseline treatment except that subjects receive individual feedback after completing the 10 Raven matrices but before stating their beliefs, the step marked “Feedback treatment only” in Figure 4. We inform them of their score and their probability of achieving Rank 1 in a five-person group, based on their score and historical performance data.

Specifically, we calculate this probability by comparing their score to a distribution of scores from Baseline treatment subjects. For each score, we bootstrap 10,000 five-person groups and calculate how often that score is the best in the five-person group, with random tie-breaking. The probability for achieving the best score in the group is the proportion of times the score ranks the best across these simulations. The Rank-1 probability is the probability for having the best score multiplied by the probability that the rank will be determined by performance (80%), in addition to the probability for being in Rank 1 by luck ($20\% * 20\% = 4\%$). Subjects can access detailed information about this calculation through an optional button.

The remainder of Period 1 proceeds as in the Baseline treatment. After receiving the feedback, subjects state their beliefs about achieving each rank in the group. Like the Baseline treatment, there is a High-inequality condition and a Low-inequality condition. By providing accurate performance feedback, we expect to observe a smaller effect of Period 1 inequality on confidence and thus Period 2 inequality choice.

Implementation and sample. Our sample consists of US nationals recruited via the crowd-sourcing web service Prolific. Data collection was done in November 2024. On average, participants earned 5 USD and spent 22 minutes on the study. There are $N=773$ complete submissions in the baseline treatment and $N=426$ observations in the feedback treatment. The experiment was programmed using oTree (Chen et al., 2016).

5 Experimental results

The analyses in this section were pre-specified in the pre-analysis plan unless indicated otherwise. Descriptive statistics and balance tests are in Appendix Table A.2. There are $N = 377$ subjects in the High inequality condition and $N = 396$ in the Low inequality condition, and the balance tests reveal no significant differences in socio-demographic characteristics.

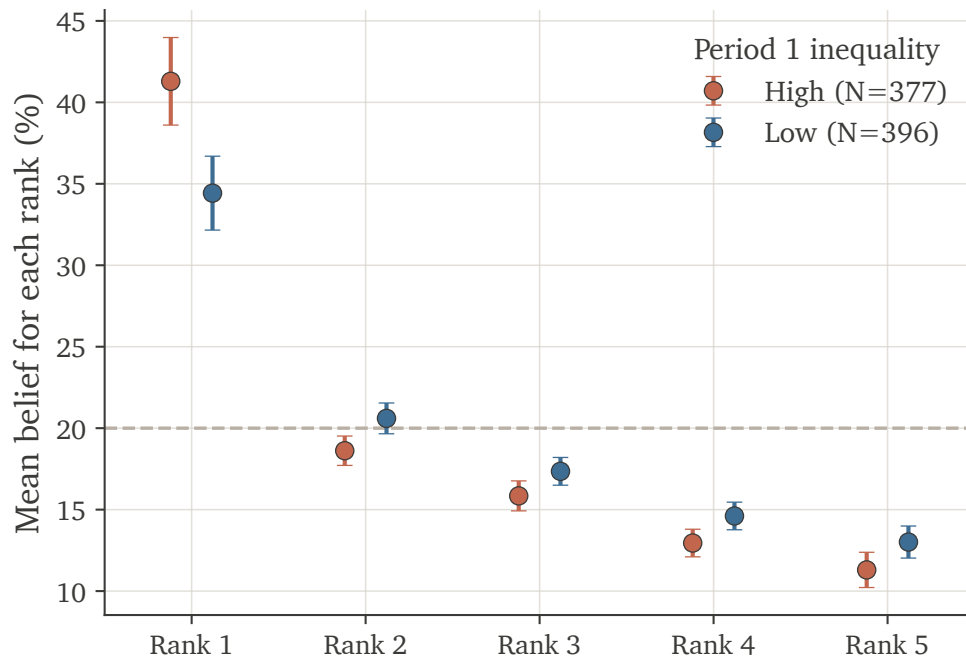
5.1 Assigned inequality and beliefs

Figure 5 plots average beliefs for each rank by inequality condition. Two features stand out. First, subjects are overconfident about the top. Regardless of their assigned inequality, they believe their chance of obtaining Rank 1 is well above the 20% group average, while their

average belief about Rank 2 sits close to 20%. That subjects inflate their belief about Rank 1 but not Rank 2 is itself consistent with motivated reasoning: Rank 1 pays more than every other rank, whereas Rank 2 pays the same as Ranks 3 through 5.

Second, and central to our prediction, the Rank-1 belief responds strongly to inequality. Subjects in the High inequality condition report an average Rank-1 belief of 41.3%, against 34.4% in the Low inequality condition, a 6.9 percentage point gap that is significant at the 1% level (Column 1 of Appendix Table A.4). The objective probability is 20% in both conditions, so the gap cannot reflect any real difference in the chance of becoming Rich. It is only for Rank 1 that beliefs rise with inequality; for every other rank, average beliefs are if anything lower in the High condition.

Figure 5: Average beliefs for each rank by inequality condition



Notes: Subjects' average belief for each rank (1–5) by inequality condition. Error bars are 95% confidence intervals and the dashed line marks the 20% group-average probability. Only the Rank-1 belief rises with inequality; beliefs about all other ranks are flat or lower. The full distribution of Rank-1 and Rank-2 beliefs is in Appendix Figure A.3.

Robustness. The shift is in the full distribution, not just the mean: a Kolmogorov-Smirnov test rejects equality of the Rank-1 belief distributions ($p < 0.01$), and the share reporting a strictly above-average chance (above 20%) rises from 54.3% under Low inequality to 64.7% under High inequality (Appendix Figure A.3). Because rank is determined by performance with 80% probability and by luck with 20%, the objective Rank-1 probability can never exceed 84%; only 38 of 773 subjects report a probability above this ceiling, almost all for Rank 1 and disproportionately under High inequality ($p < 0.05$), which points to motivated distortion rather than confusion. Excluding or censoring these subjects leaves the treatment effect unchanged (Appendix Table A.5).

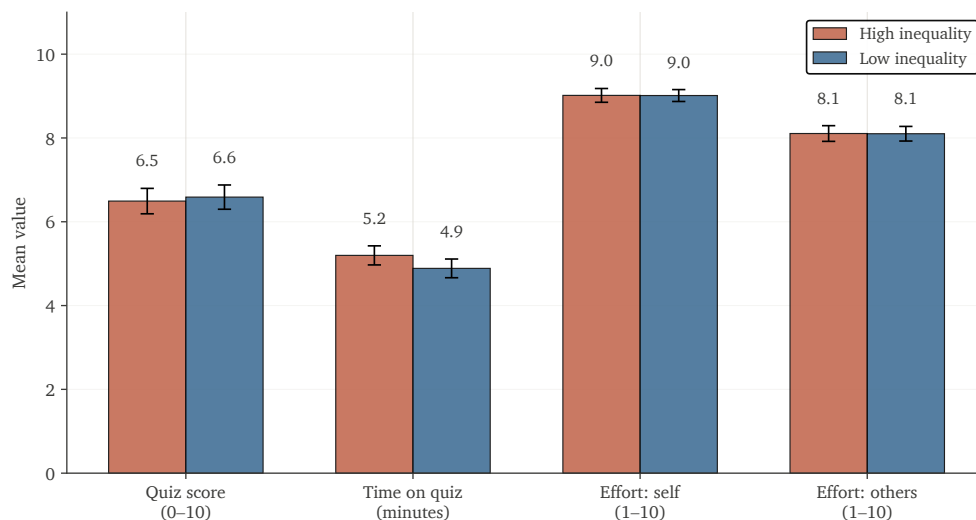
Result 4: *Experiencing higher initial inequality causes higher overconfidence about obtaining the top rank.*

5.2 Is the overoptimism motivated?

A natural alternative to motivated beliefs is performance optimism: subjects in the High condition might simply believe they performed better on the quiz. We rule this out in two steps.

First, there is no difference in performance or effort across conditions. Figure 6 shows that quiz scores, time on the quiz, self-reported effort, and beliefs about others' effort are all statistically indistinguishable between conditions. This also rules out equilibrium neglect, the possibility that subjects respond to the stronger incentive under High inequality by working harder while failing to anticipate that others do the same: there is no extra effort to neglect.

Figure 6: Performance and effort do not differ across conditions



Notes: Mean quiz score (out of 10), time on the quiz (maximum time allowed: 10 minutes), self-reported own effort from 1 to 10, and perceived effort of others from 1 to 10, by inequality condition. Error bars are 95% confidence intervals.

Second, beliefs about Rank 2 provide a sharp test that turns the two accounts against each other. Performance optimism and motivated beliefs make opposite predictions here. Since Rank 1 and Rank 2 are both above-median outcomes in a five-person group, a subject who believes they performed better should raise the probability they assign to *both* ranks. Motivated beliefs, by contrast, target only the high-paying position: because probabilities must sum to 100, inflating Rank 1 should leave Rank 2 flat or push it down. Using our empirical performance distribution, we show that any plausible belief in better performance raises both the mean and the median Rank-2 belief, and we prove that the median must rise for any performance distribution unless the improvement is implausibly large (Appendix E.1). The data show the opposite (Figure 5): as Rank-1 beliefs rise under High inequality, Rank-2 beliefs stay flat at the median (20.0% versus 20.0%) and fall at the mean (18.6%

versus 20.6%, $p < 0.01$). The increase appears only for the high-paying position, which is the signature of motivated belief distortion rather than optimism about performance.

Robustness. The Rank-2 falsification is general, not an artifact of our sample: we prove the median Rank-2 belief must rise under any performance-optimism account for any performance distribution unless the improvement in believed performance is implausibly large (Appendix E.1), yet the data show the opposite. The result is also not driven by differential effort, since quiz score, time on the quiz, and self-reported effort do not differ across conditions (Figure 6).

Result 5: *The inequality-induced overoptimism is motivated, not performance-based: it appears only for the high-paying rank, with no difference in performance, effort, or beliefs about other above-median ranks.*

5.3 Robustness to accurate feedback

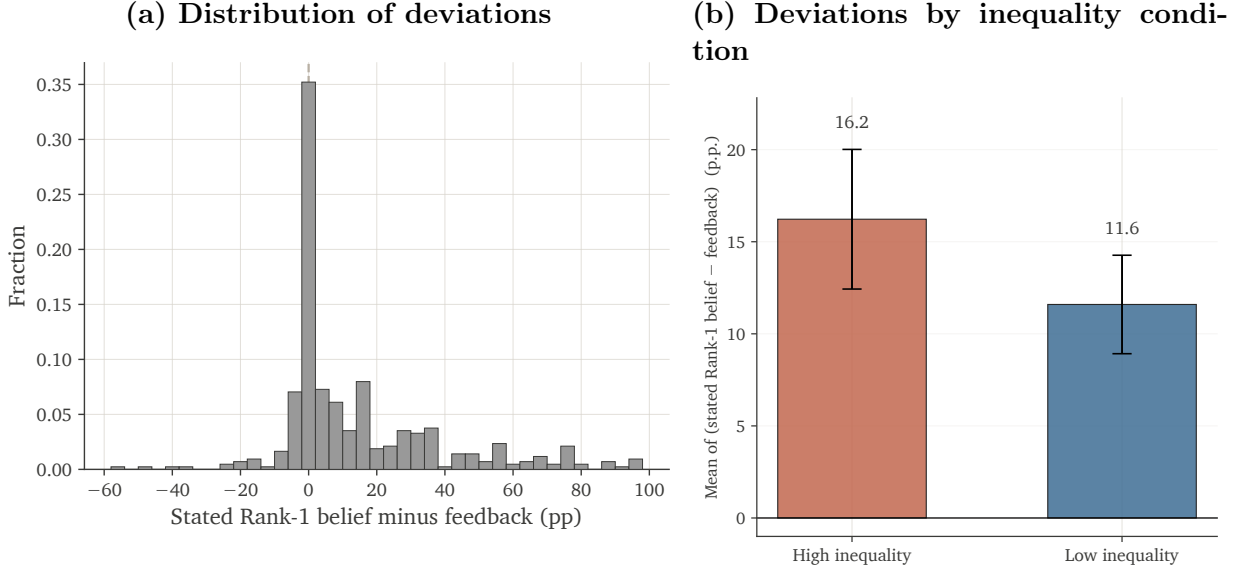
If the overconfidence reflected missing information rather than motivated reasoning, telling subjects their true chance of becoming Rich should erase the inequality gap. In the Feedback treatment, subjects learn their quiz score and their objective Rank-1 probability before reporting beliefs (Section 4). Feedback does move beliefs: average Rank-1 beliefs fall from 38.8% in the Baseline to 31.4% under Feedback ($p < 0.01$), and 30% of subjects report exactly the probability they were given. But the inequality gap persists. Subjects in the High condition still report higher Rank-1 beliefs than those in the Low condition (33.7% versus 29.3%, $p < 0.10$) even after receiving the correct number; the treatment effect on beliefs shrinks relative to the Baseline but remains positive (Appendix Figure A.2).

Because Feedback subjects are told their objective probability, we can measure belief distortion directly as the gap between the reported belief and the probability provided. Figure 7 shows two things. Panel (a): deviations are sharply asymmetric. Beyond the 30% who follow the feedback exactly, subjects are far more likely to report a belief above their given probability than below it. Panel (b): the upward deviation is larger under High inequality. Even though every subject was told their accurate, individualized probability, those in the High condition inflate their stated belief above the feedback by 16.2 percentage points on average, more than under Low inequality (Figure 7). The persistence of the gap, and the fact that it is larger exactly where the motive to distort is strongest, is what motivated reasoning predicts and what missing information cannot explain.

Robustness. The residual upward distortion is concentrated among subjects who scored below the median, exactly where the motive to distort is strongest, and the High-Low gap there is significant at the 5% level (Appendix Figure A.4).

Result 6: *The inequality-induced overconfidence persists despite accurate, individualized feedback, and the residual distortion is largest among those with the strongest motive to hold optimistic beliefs.*

Figure 7: Deviations of Rank-1 beliefs from the feedback provided



Notes: Deviation of the stated Rank-1 belief from the objective probability provided in the feedback; positive values indicate overconfidence relative to the feedback. Panel (a): full distribution of deviations. Panel (b): mean deviation by inequality condition; the High condition inflates beliefs above the feedback more than the Low condition (16.2 versus 11.6 percentage points). Error bars are 95% confidence intervals.

5.4 Initial inequality and inequality acceptance

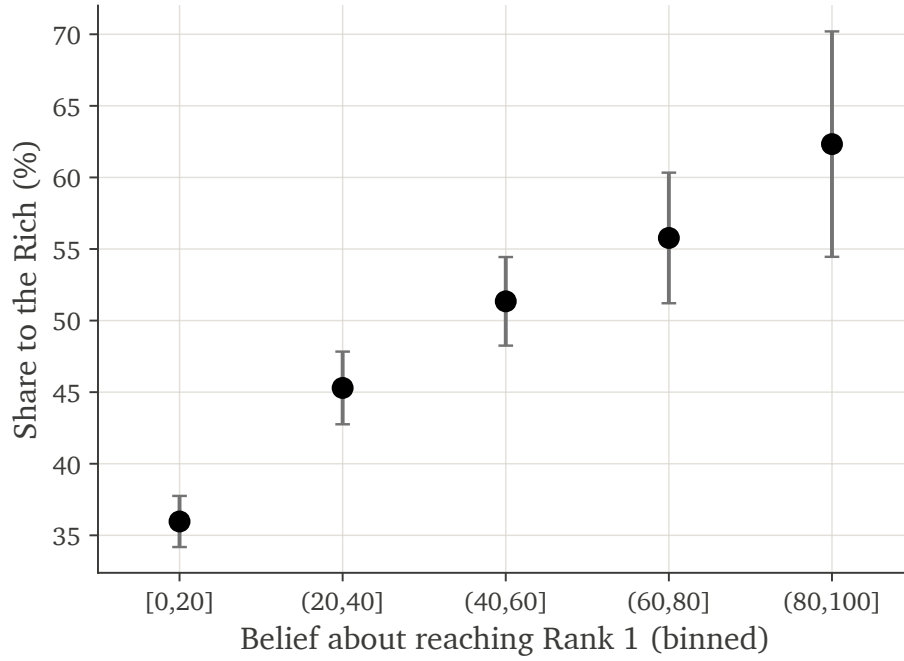
We now show that this belief shift carries through to behavior. In Period 2, before learning their rank, subjects choose how much of \$10 to allocate to the Rich, with the rest split equally among the four Poor and one subject's choice implemented for the group. Figure 8 relates this choice to the Rank-1 belief: the share allocated to the Rich rises monotonically with confidence about becoming Rich, from about 35% among the least confident to about 60% among the most confident. Subjects prefer more inequality precisely when they expect to be the one who benefits from it, the logic at the heart of the prospect of upward mobility hypothesis.

Because higher inequality raises the Rank-1 belief, and the Rank-1 belief raises the share to the Rich, higher inequality should raise inequality acceptance. It does. Subjects in the High condition allocate 48.0% to the Rich on average, against 40.8% in the Low condition, a 7.2 percentage point gap significant at the 1% level (Column 3 of Appendix Table A.4). The inequality subjects are handed in Period 1 becomes the inequality they choose for Period 2.

Robustness. The gap is in the full allocation distributions, not just the mean: a Kolmogorov-Smirnov test rejects equality of the distributions ($p < 0.01$; Appendix Figure A.1), and the share of subjects who fully equalize income is higher under Low inequality (27.0% versus 21.0%, $p < 0.05$). The treatment effect is also robust to demographic controls (Appendix Table A.4).

Result 7: *Experiencing higher initial inequality causes higher inequality acceptance.*

Figure 8: Share to the Rich by Rank-1 belief



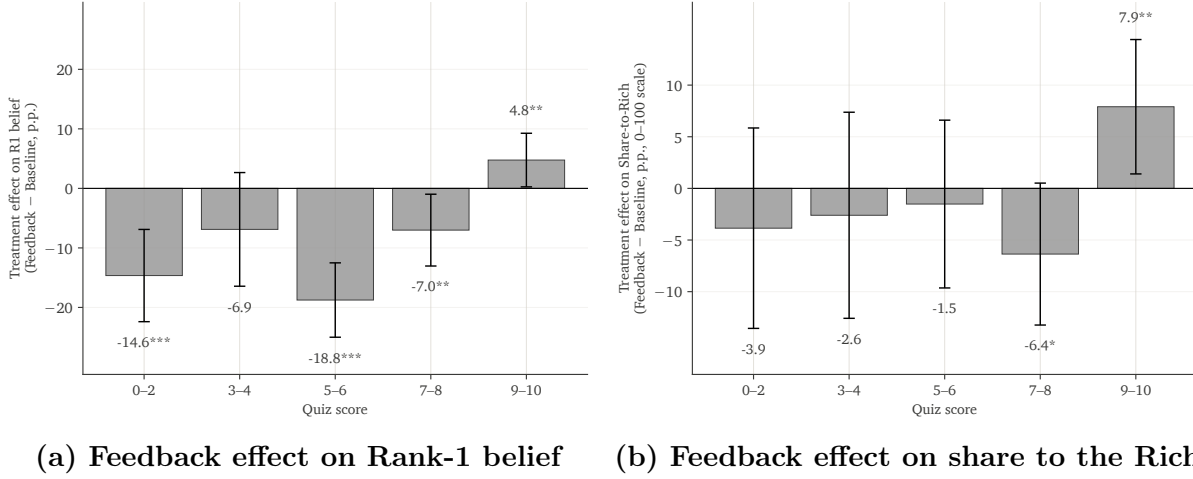
Notes: Average share allocated to the Rich by Rank-1 belief, pooling both conditions. Rank-1 beliefs are grouped into five bins. Error bars are 95% confidence intervals.

5.5 The belief channel is causal

The treatment effect on inequality acceptance is consistent with the belief channel, but inequality might also raise the share to the Rich directly, for instance by shifting fairness views (Cappelen et al., 2007; Almås et al., 2020) or anchoring. To isolate the belief channel, we use the Feedback treatment as a source of *exogenous* variation in beliefs. The feedback is individualized: it tells each subject their own objective probability, which is low for weak performers and high for strong ones. It therefore moves beliefs in different directions depending on performance, pushing down the optimism of those who overestimated themselves and confirming or raising the beliefs of those who did well. This gives us an instrument for beliefs that is driven by the feedback, not by the subject's fairness preferences or other tastes that might independently affect the allocation.

Figure 9 makes the logic visible before any regression. Panel (a) is the first stage: it plots how feedback shifts the Rank-1 belief, by quiz score. For low and middle scorers, feedback lowers the belief sharply (by 15 to 19 percentage points); for the top scorers, feedback *raises* it (by 4.8 points), because they learn their true chance is higher than they had assumed. Panel (b) is the reduced form: the effect of the same feedback on the share to the Rich, by the same score bins. The two panels track each other. Where feedback pulls beliefs down, the share to the Rich is flat to negative; where feedback pushes beliefs up, among the top scorers, the share to the Rich rises in lockstep, by 7.9 percentage points. The behavior follows the belief wherever the feedback moves it.

Figure 9: Feedback as an instrument: belief and behavior move together



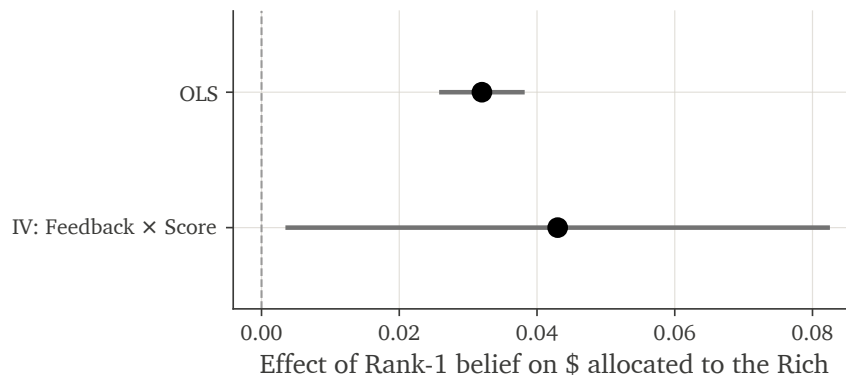
Notes: Treatment effect of Feedback (Feedback minus Baseline) by quiz-score bin. Panel (a): effect on the Rank-1 belief (first stage). Panel (b): effect on the share allocated to the Rich (reduced form). Feedback lowers beliefs for low and middle scorers and raises them for top scorers; the share to the Rich moves in the same direction wherever the belief moves. Error bars are 95% confidence intervals; stars denote significance at the 10/5/1% level.

We formalize this with two-stage least squares (Figure 10; full estimates in Appendix Table A.1), instrumenting the Rank-1 belief with the interaction of feedback and quiz score and controlling for feedback, quiz score, its square, and the inequality condition. Feedback on its own is a weak instrument, because it raises some subjects' beliefs and lowers others' so its average effect on beliefs is small; the performance-interacted instrument captures this heterogeneous shift and is strong, with a first-stage F of 23.1. The instrumented effect of the Rank-1 belief on the share to the Rich is 0.043 (SE 0.020, $p < 0.05$): a 10 percentage point increase in the Rank-1 belief raises the amount allocated to the Rich by about \$0.43 out of \$10, equivalently 4.3 percentage points of the share. This is, if anything, slightly larger than the OLS association of 0.032, so confounding does not inflate the link.¹⁵

Robustness. The identifying assumption is that feedback affects the allocation only through beliefs. The natural threat is that learning one's score also makes performance, and hence desert, more salient when choosing. Because the estimate is identified off the *interaction* of feedback with score rather than the level of feedback, any uniform effect of receiving feedback, including a general rise in the salience of performance or desert, is absorbed by the feedback main effect; what identification requires is only that feedback not shift the performance-to-allocation mapping differentially by score through some channel other than beliefs. The estimate is a local average treatment effect for the subjects whose beliefs respond to feedback, which is the relevant population for the belief channel. We stress that beliefs are *a* causal channel, not the entire mechanism: the inequality treatment effect on the share to the Rich remains positive and significant after controlling for the Rank-1 belief (column 4 of Appendix Table A.4), so inequality also operates through other channels, such as fairness

¹⁵An over-identified specification that adds feedback as a second instrument does not reject the exclusion restriction (Hansen-Sargan).

Figure 10: The belief channel: OLS and instrumental-variables estimates



Notes: Effect of the Rank-1 belief (0 to 100) on the dollar amount allocated to the Rich (0 to 10), with 95% confidence intervals. OLS and two-stage least squares instrumenting the Rank-1 belief with the interaction of feedback and quiz score (first-stage $F = 23.1$), controlling for the inequality condition, quiz score, and its square. Full estimates in Appendix Table A.1.

views anchored by the initial distribution. What the instrument establishes is that the belief channel is genuinely causal rather than a self-selection artifact.

Result 8: *The effect of inequality on inequality acceptance runs causally through beliefs. Instrumenting the Rank-1 belief with individualized feedback yields a positive, significant effect of beliefs on the share allocated to the Rich.*

5.6 Heterogeneity analysis

We examine heterogeneity in the treatment effects by gender, income, age, and political orientation (Appendix Figures A.8 and Tables A.7 and A.8). Baseline confidence and inequality acceptance are higher among men than women and among Republicans than Democrats, but no interaction of any subgroup with the High inequality condition is significant at the 5% level. The direction of the effect is the same in every subgroup: High inequality raises Rank-1 beliefs and the share to the Rich. While the power to detect interactions is limited, the consistency of the effect indicates that inequality-induced motivated beliefs operate across diverse subpopulations rather than being driven by any one group.

6 Conclusion

This paper proposes and tests a resolution to a long-standing puzzle. The prospect of upward mobility can reconcile high inequality with weak demand for redistribution only if citizens of unequal societies are more optimistic about moving up, yet the Great Gatsby Curve shows that these are exactly the societies where actual mobility is lowest. We argue that inequality itself closes this gap. Because a more concentrated top raises the anticipatory utility of believing one will reach it, higher inequality manufactures more optimistic beliefs about upward mobility.

The cross-country survey evidence shows the pattern in the field. Citizens of more unequal countries expect more upward mobility even as they report less actual mobility, a belief reversal of the Great Gatsby Curve. The same holds within countries: individuals who experienced higher inequality during their formative years expect more upward mobility decades later, and this optimism concentrates on the destinations that pay above the mean income rather than rising uniformly across the ladder. Where citizens expect higher future status, they demand less redistribution, the association the prospect of upward mobility hypothesis requires.

The experiment supplies the causal core that the survey cannot. Participants randomly assigned to a more unequal society become more confident about reaching the top, and this confidence is motivated rather than a belief about better performance: it appears only for the high-paying position, with no difference in performance, effort, or beliefs about other above-median ranks, and it persists even when participants are told their accurate, individualized chance of success. The induced optimism in turn raises the inequality participants accept. Using the individualized feedback as an instrument for beliefs, we show that this effect runs causally through the belief channel, though beliefs are one channel among others rather than the whole of it.

The two halves of the paper point to the same conclusion: inequality raises optimism about future status, and that optimism lowers support for redistribution. This offers a behavioral account of why rising inequality need not generate the demand for redistribution that the median-voter logic predicts. When inequality distorts beliefs about who will benefit from it, it can blunt its own political correction, which is one way inequality may sustain itself.

The same beliefs that dampen redistribution may also provide felicity to those who hold them. Using US data from the General Social Survey, we find that more optimistic expectations about future status are correlated with higher current happiness, controlling for current status, and that this correlation is concentrated among low-income respondents. We read this as suggestive rather than causal, but it points to a genuine welfare tension: the optimism that helps sustain inequality may also be a source of well-being for precisely those whom redistribution would most help. Correcting these beliefs could improve democratic deliberation while lowering the felicity of the disadvantaged, a trade-off that any informational policy must weigh.

Finally, our mechanism reframes the American Dream. Critics read declining mobility as evidence that the dream is broken, while defenders insist it survives. Our findings suggest that the dream's force lies less in its accuracy than in its anticipatory value, and that this value rises precisely as inequality rises and actual mobility falls. This may help explain why optimism about mobility is most pronounced in the most unequal societies. It need not reflect ignorance; it can be motivated belief, shaped by the very stakes that make reaching the top both more rewarding to imagine and less likely to attain.

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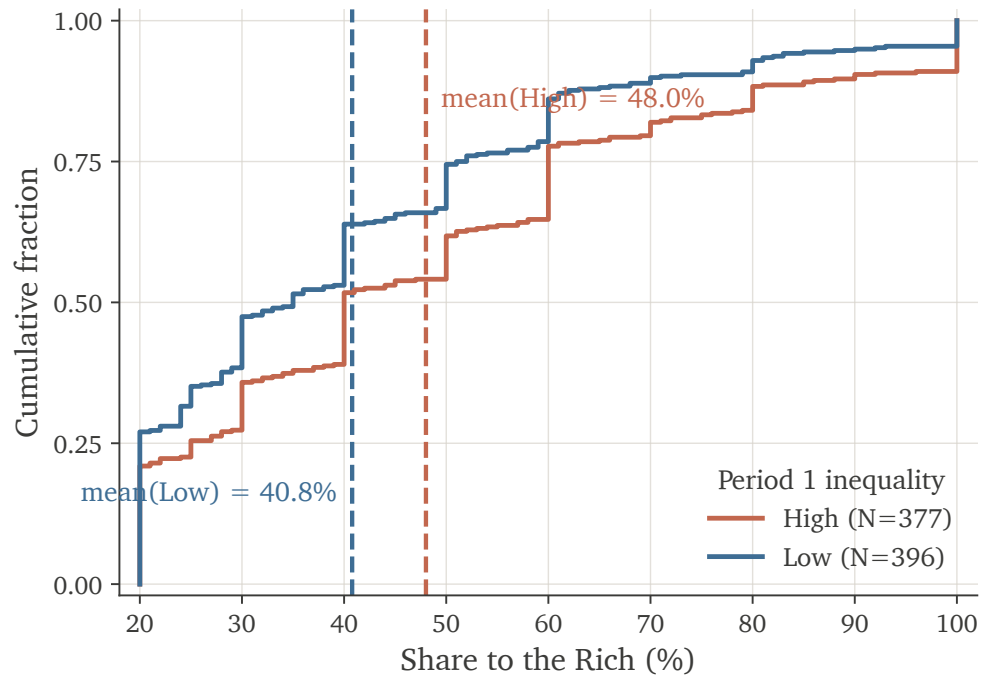
A Additional figures and tables - experimental evidence

Table A.1: The belief channel: OLS and instrumental-variables estimates

	(1) OLS	(2) IV: Feedback $\times$ Score
Rank-1 belief	0.032*** (0.003)	0.043** (0.020)
First-stage F		23.1
N	1,199	1,199

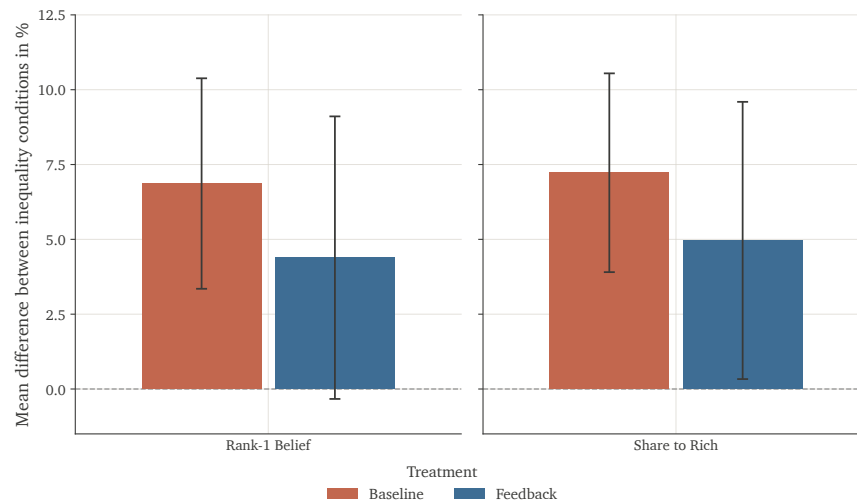
Notes: The outcome is the dollar amount allocated to the Rich (0–10); the endogenous regressor is the Rank-1 belief (0–100). Column (1) is OLS. Column (2) is two-stage least squares instrumenting the Rank-1 belief with the interaction of feedback and quiz score, controlling for the inequality condition, quiz score, and its square. Feedback alone is a weak instrument, because its average effect on beliefs is small (it raises some subjects' beliefs and lowers others'); the performance-interacted instrument captures this heterogeneous shift and is strong. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.1: Share to the Rich by inequality condition



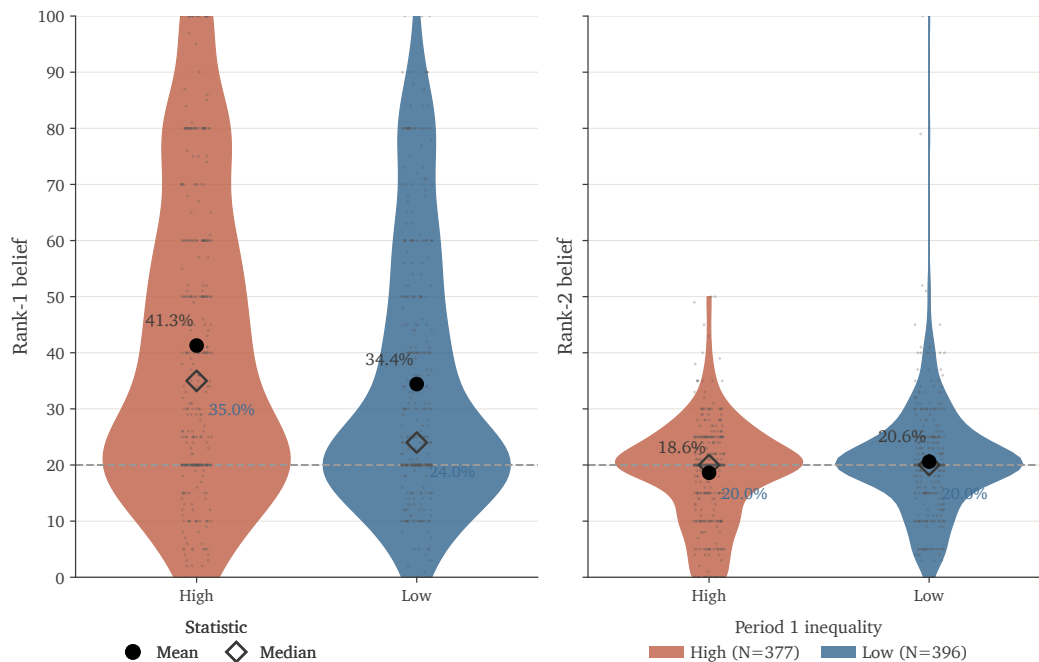
Notes: Cumulative distribution of the share allocated to the Rich by inequality condition. The red line is the High inequality condition, the blue line the Low. Vertical dashed lines mark condition means. A Kolmogorov-Smirnov test rejects equality of the two distributions ($p < 0.01$).

Figure A.2: Treatment effect on beliefs and inequality acceptance: Baseline versus Feedback



Notes: Mean difference between inequality conditions (High minus Low, in percentage points) for the Rank-1 belief (left) and the share to the Rich (right), by treatment. The inequality gap falls under Feedback but remains positive; the difference between the Baseline and Feedback treatment effects is not statistically significant. Error bars are 95% confidence intervals.

Figure A.3: Distribution of Rank-1 and Rank-2 beliefs by inequality condition



Notes: Distribution of Rank-1 and Rank-2 beliefs by inequality condition (Baseline treatment). Black dots are means, blue triangles medians. A Kolmogorov-Smirnov test rejects equality of the Rank-1 belief distributions across conditions ($p < 0.01$); the High distribution has more mass in the upper tail. Companion to Figure 5.

Table A.2: Descriptive statistics by inequality condition (Baseline treatment)

Treatment Variable	High Inequality			Low Inequality			Test
	N	Mean	SD	N	Mean	SD	
Gender	377			396			X2=1.285
... Female	186	49%		198	50%		
... Male	186	49%		190	48%		
... Other	5	1%		7	2%		
... Rather not say	0	0%		1	0%		
Age in years	377	38	13	396	37	12	F=0.709
College degree (0/1)	377	0.7	0.46	396	0.7	0.46	F=0.01
Ethnicity	377			396			X2=4.159
... White	226	60%		234	59%		
... African American	78	21%		101	26%		
... Hispanic	18	5%		14	4%		
... East Asian	22	6%		19	5%		
... Other	30	8%		24	6%		F=1.205
... Prefer not to say	3	1%		4	1%		
Voted Republican in last election (0/1)	377	0.32	0.47	396	0.36	0.48	
Household income last year (USD)	377			396			X2=3.088
... \$0 - \$30,000	61	16%		76	19%		
... \$30,001 - \$60,000	93	25%		106	27%		
... \$60,001 - \$100,000	99	26%		96	24%		
... \$100,001 - \$150,000	74	20%		75	19%		
... \$150,001 or above	39	10%		36	9%		F=14.748***
... Prefer not to say	11	3%		7	2%		
Rank-1 belief	377	41	27	396	34	23	
Chosen income share to the Rich (Rank 1)	377	48	25	396	41	22	
Score on quiz	377	6.5	3	396	6.6	2.9	F=0.199
Time for quiz (max. 10 minutes)	377	5.2	2.2	396	4.9	2.3	F=3.653
Importance of effort for success in quiz (0-100)	377	41	18	396	40	16	F=1.648
Importance of luck for success in quiz (0-100)	377	18	14	396	18	13	F=0.005
Importance of intelligence for success in quiz (0-100)	377	41	19	396	43	17	F=1.358
Self-reported effort on quiz (0-10)	377	9	1.6	396	9	1.4	F=0.001
Belief about effort of others on quiz (0-10)	377	8.1	1.8	396	8.1	1.8	F=0.002
Willingness to take risks (1-10)	377	6	2.6	396	5.9	2.6	F=0.028
It is fair for Rank 1 to earn more (0-10)	377	7	2.6	396	7	2.6	F=0.001
Society should aim to equalize incomes (0-10)	377	5.2	2.7	396	5.2	2.8	F=0
Position in relative income distribution (1-5)	377	2.7	0.73	396	2.8	0.71	F=0.057
Chance for being in highest income group (0-100)	377	9.3	15	396	8.6	14	F=0.533
Used genAI tools	377	2	0.17	396	2	0.1	F=0.084
Thought others used genAI	377	1.9	0.35	396	1.9	0.33	F=0.478

Statistical significance markers: * p<0.05; ** p<0.01; *** p<0.001

Table A.3: Empirical distribution of quiz scores and rank probabilities

Score	Share (%)	P(Rank 1)	P(Rank 2)	P(Rank 3)	P(Rank 4)	P(Rank 5)
0	2.92	4.0%	4.0%	4.1%	8.3%	79.6%
1	4.50	4.0%	4.0%	5.1%	18.2%	68.7%
2	5.42	4.0%	4.3%	8.1%	28.0%	55.6%
3	6.26	4.1%	5.1%	12.6%	34.1%	44.2%
4	6.17	4.1%	6.4%	18.4%	37.3%	33.7%
5	8.76	4.7%	10.0%	25.4%	36.5%	23.4%
6	9.17	5.8%	15.9%	30.7%	32.5%	15.2%
7	9.67	8.3%	22.9%	33.4%	25.6%	9.8%
8	13.93	14.6%	31.0%	31.9%	16.6%	5.8%
9	15.51	28.9%	37.2%	21.2%	8.2%	4.5%
10	17.68	59.8%	24.0%	7.8%	4.3%	4.0%
Average		20.0%	20.0%	20.0%	20.0%	20.0%

Table A.4: Treatment effects on inequality acceptance and Rank-1 belief

	(1) Rank-1 belief	(2) Share to Rich	(3) Share to Rich	(4) Share to Rich
High Period 1 inequality	6.865*** (1.788)		7.225*** (1.689)	5.305** (1.629)
Rank-1 belief		0.294*** (0.032)		0.280*** (0.033)
Constant	34.427*** (1.248)	33.198*** (1.469)	40.788*** (1.179)	31.158*** (1.589)
Num.Obs.	773	773	773	773
R2	0.019	0.097	0.023	0.109

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

OLS regression estimates of treatment effects on inequality acceptance (share allocated to Rich) and Rank-1 belief. Column (1) reports coefficients for Rank-1 belief; columns (2)-(4) report coefficients for the share allocated to the rich. Standard errors are reported in parentheses. Sample size and R² are shown for each specification.

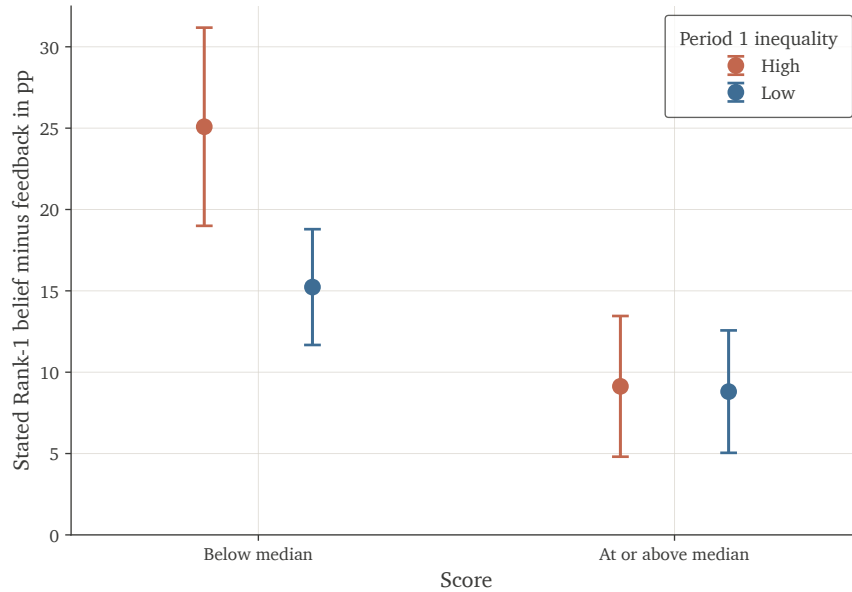
Table A.5: Treatment effect - Robustness check for sample with rank probabilities $> 84\%$

	(1) Rank-1 belief	(2) Rank-1 belief	(3) Rank-1 belief	(4) Share to Rich	(5) Share to Rich	(6) Share to Rich
High Period 1 inequality	6.865*** (1.788)	4.539** (1.613)	6.238*** (1.681)			
Rank-1 belief				0.294*** (0.032)	0.312*** (0.037)	0.314*** (0.034)
Constant	34.427*** (1.248)	32.859*** (1.115)	34.035*** (1.174)	33.198*** (1.469)	32.584*** (1.531)	32.676*** (1.514)
Num.Obs.	773	735	773	773	735	773
R2	0.019	0.011	0.018	0.097	0.088	0.097

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

OLS regression estimates of treatment effects on Rank-1 belief (1-3) and effect of Rank-1 belief on Share to rich (4-6). Columns (1) and (4) show baseline results. Columns (2) and (4) restrict the sample to those who did not state impossible probabilities (> 84 percent). Columns (3) and (6) show results when replacing stated Rank-1 probabilities by the formula (stated prob.) $\times 0.8 + 4 \times 0.2$, i.e. to include luck. Standard errors are reported in parentheses. Sample size and R^2 are shown for each specification.

Figure A.4: Deviations from feedback by performance and inequality condition



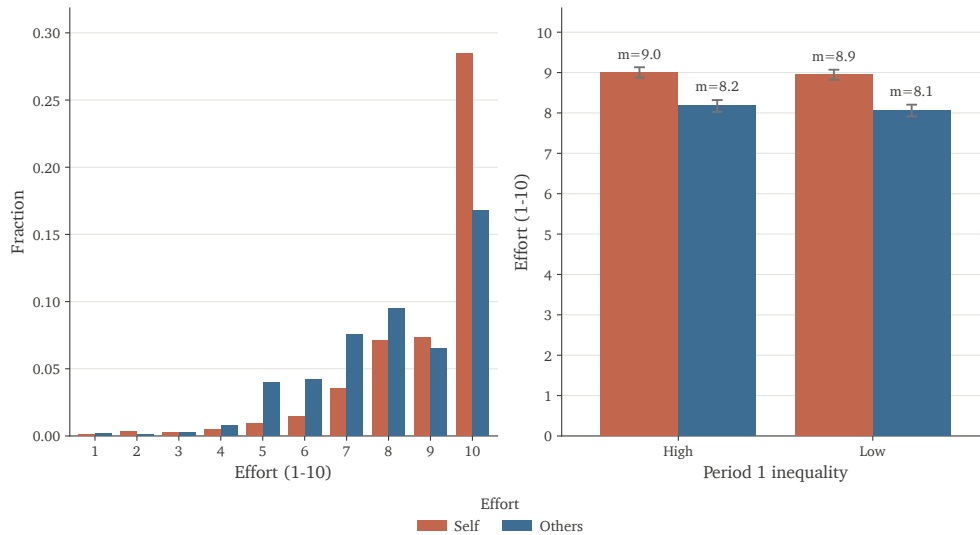
Notes: Mean deviation of the stated Rank-1 belief from the feedback provided, by inequality condition and performance. “Below median” is a quiz score of 6 or fewer correct answers. The residual overconfidence is concentrated among below-median performers, among whom the High condition deviates significantly more than the Low ($p < 0.05$); among above-median performers the conditions do not differ. Error bars are 95% confidence intervals.

Table A.6: Descriptive statistics by inequality condition (Feedback treatment)

Treatment Variable	High Inequality			Low Inequality			Test
	N	Mean	SD	N	Mean	SD	
Gender	207			219			X2=0.568
... Female	97	47%		102	47%		
... Male	108	52%		113	52%		
... Other	2	1%		4	2%		
Age in years	207	38	12	219	39	13	F=0.969
College degree (0/1)	207	0.65	0.48	219	0.72	0.45	F=2.38
Ethnicity	207			219			X2=2.566
... White	137	66%		139	63%		
... African American	45	22%		52	24%		
... Hispanic	7	3%		12	5%		
... East Asian	4	2%		6	3%		
... Other	14	7%		10	5%		
... Prefer not to say	0	0%		0	0%		
Voted Republican in last election (0/1)	207	0.3	0.46	219	0.4	0.49	F=4.496*
Household income last year (USD)	207			219			X2=5.009
... \$0 - \$30,000	36	17%		25	11%		
... \$30,001 - \$60,000	61	29%		58	26%		
... \$60,001 - \$100,000	54	26%		64	29%		
... \$100,001 - \$150,000	35	17%		42	19%		
... \$150,001 or above	18	9%		27	12%		
... Prefer not to say	3	1%		3	1%		
Rank-1 belief	207	34	27	219	29	22	F=3.364
Chosen income share to the Rich (Rank 1)	207	47	25	219	42	23	F=4.436*
Score on quiz	207	6.5	2.9	219	6.5	2.9	F=0.034
Time for quiz (max. 10 minutes)	207	5.3	2.3	219	5	2.1	F=2.389
Importance of effort for success in quiz (0-100)	207	39	16	219	40	15	F=0.133
Importance of luck for success in quiz (0-100)	207	20	15	219	20	15	F=0.009
Importance of intelligence for success in quiz (0-100)	207	41	18	219	40	16	F=0.177
Self-reported effort on quiz (0-10)	207	9	1.5	219	8.8	1.8	F=0.917
Belief about effort of others on quiz (0-10)	207	8.3	1.7	219	8	2	F=2.976
Willingness to take risks (1-10)	207	5.7	2.6	219	6.2	2.5	F=3.396
It is fair for Rank 1 to earn more (0-10)	207	7.2	2.6	219	7	2.6	F=0.343
Society should aim to equalize incomes (0-10)	207	4.9	2.8	219	5.3	2.8	F=1.589
Position in relative income distribution (1-5)	207	2.7	0.75	219	2.8	0.65	F=1.41
Chance for being in highest income group (0-100)	207	8.2	16	219	8.9	15	F=0.235
Used genAI tools	207	2	0.18	219	2	0.095	F=1.149
Thought others used genAI	207	1.9	0.33	219	1.9	0.35	F=0.588

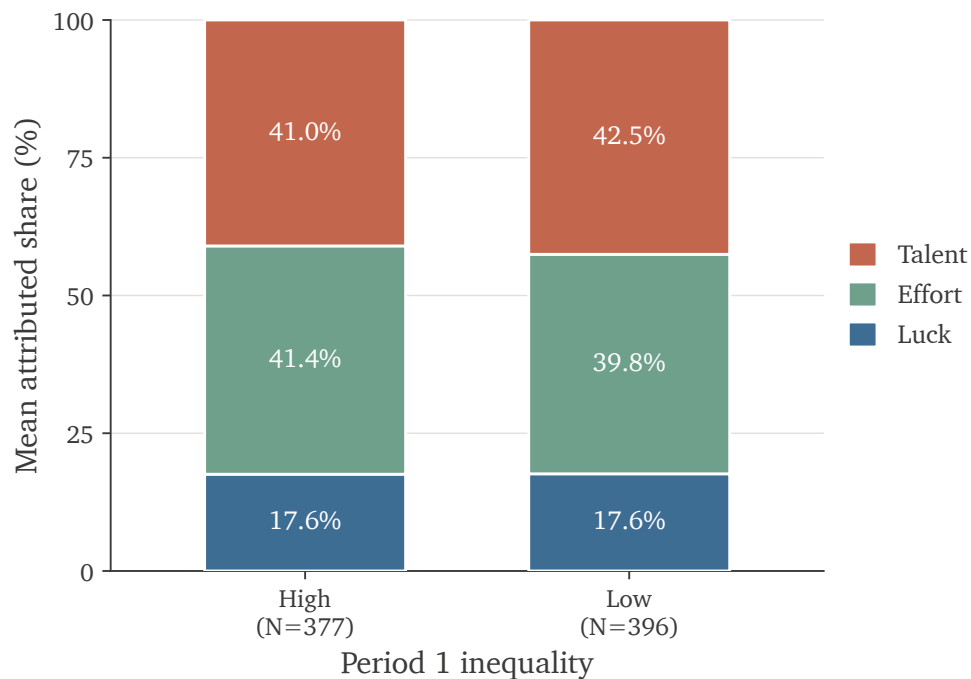
Statistical significance markers: * p<0.05; ** p<0.01; *** p<0.001

Figure A.5: Survey evidence: self-reported effort and estimated effort of others



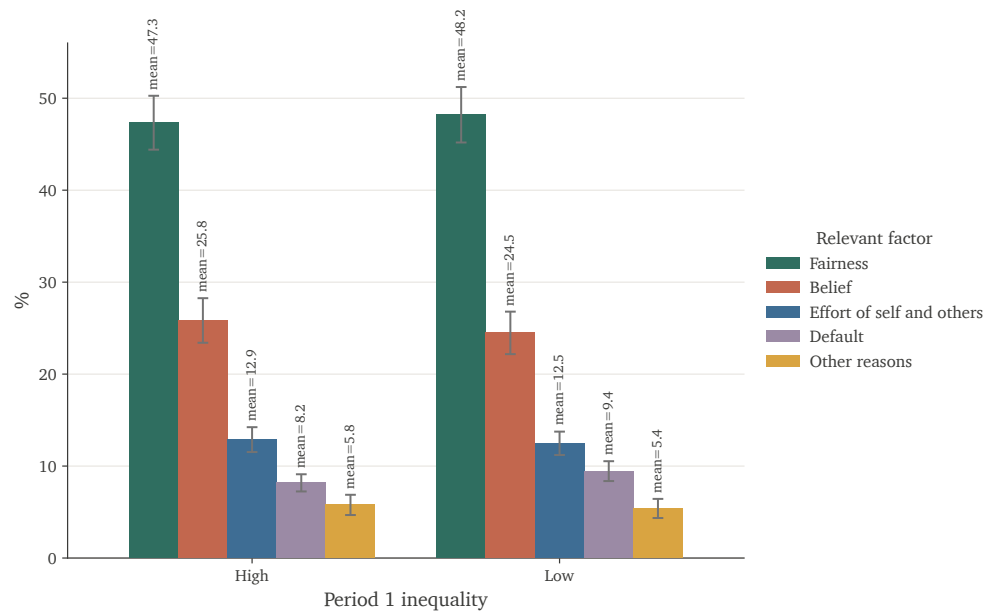
Notes: The figure depicts how much effort participants in the experiment reported to have exerted effort in the quiz task and how much effort they thought others exerted, both on a scale 0-10. The right side panel splits up these answers by inequality condition.

Figure A.6: Survey evidence: Factors determining success in the quiz task



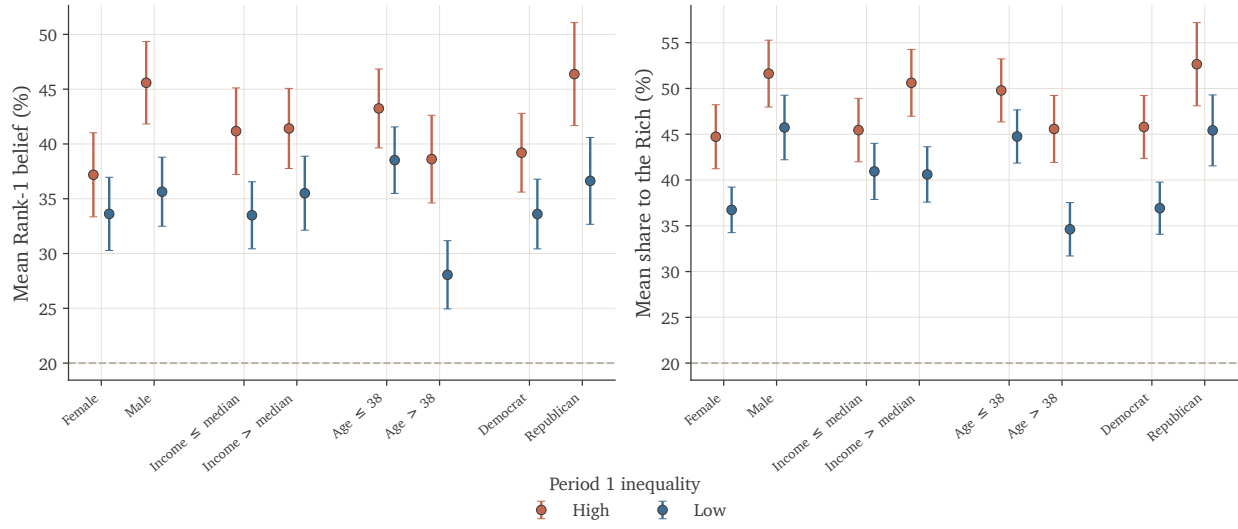
Notes: The figure depicts the average shares of effort, talent and luck the participants in the experiment thought were needed to succeed in the quiz task, by inequality condition.

Figure A.7: Survey evidence: relevant factors in redistribution decision



Notes: The figure depicts the factors participants reported to have influenced their allocation decision, by inequality condition. They were given 100 points to distribute between the factors fairness, belief about being Rich, effort, the default distribution, and other factors, to explain what influenced their decision the most.

Figure A.8: Experimental evidence by subgroup



Notes: Mean Rank-1 belief (left) and mean share allocated to the Rich (right), in percent, by Period-1 inequality (High in rust, Low in slate) within each subgroup: gender, household income (split at the sample median), age (split at the US median of 38.5 years), and political affiliation. Error bars are 95% confidence intervals; the dashed line marks the 20% group-average benchmark. Baseline treatment. High inequality raises both outcomes in every subgroup, and no subgroup interaction with the High condition is significant at the 5% level (Tables A.7 and A.8).

Table A.7: Subgroup analysis: Treatment effect on Rank-1 belief

	(1)	(2)	(3)	(4)	(5)
High Period 1 inequality	6.865*** (1.788)	4.720* (2.295)	3.804 (2.479)	8.010*** (2.299)	5.677** (2.194)
Age >38		-10.465*** (2.528)			
High P1 ineq. × Age >38		5.838 (3.598)			
Male			2.336 (2.483)		
High P1 ineq. × Male			6.140+ (3.554)		
Income >median				2.802 (2.578)	
High P1 ineq. × Income >median				-3.024 (3.663)	
Republican					3.448 (2.587)
High P1 ineq. × Republican					4.071 (3.755)
Constant	34.427*** (1.248)	38.523*** (1.582)	33.306*** (1.720)	33.372*** (1.582)	33.182*** (1.554)
Num.Obs.	773	773	773	773	773
R2	0.019	0.044	0.034	0.020	0.031

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

OLS regression estimates for treatment effect of inequality on Rank-1 belief by subgroup. Interaction terms test heterogeneity by Age (>38 years; median age in the US: 38.5), Gender (Male), Income (above median; median household income in the US (2023): \$78,000), and Political Affiliation (Republican). Standard errors are in parentheses. Sample size and R² reported for each model.

Table A.8: Subgroup analysis: Treatment effect on share to Rich

	(1)	(2)	(3)	(4)	(5)
High Period 1 inequality	7.225*** (1.689)	5.034* (2.167)	8.274*** (2.322)	5.087* (2.166)	7.626*** (2.064)
Age >38		-10.136*** (2.387)			
High P1 ineq. × Age >38		5.925+ (3.398)			
Male			9.514*** (2.325)		
High P1 ineq. × Male			-2.387 (3.328)		
Income >median				0.254 (2.430)	
High P1 ineq. × Income >median				5.180 (3.452)	
Republican					7.250** (2.433)
High P1 ineq. × Republican					-0.398 (3.532)
Constant	40.788*** (1.179)	44.755*** (1.493)	36.223*** (1.611)	40.692*** (1.490)	38.170*** (1.462)
Num.Obs.	773	773	773	773	773
R2	0.023	0.049	0.055	0.029	0.043

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

OLS regression estimates for treatment effect of inequality on inequality acceptance (allocated share to Rich) by subgroup. Interaction terms test heterogeneity by Age (>38 years; median age in the US: 38.5), Gender (Male), Income (above median; median household income in the US (2023): \$78,000), and Political Attitude (Republican). Standard errors are in parentheses. Sample size and R² reported for each model.

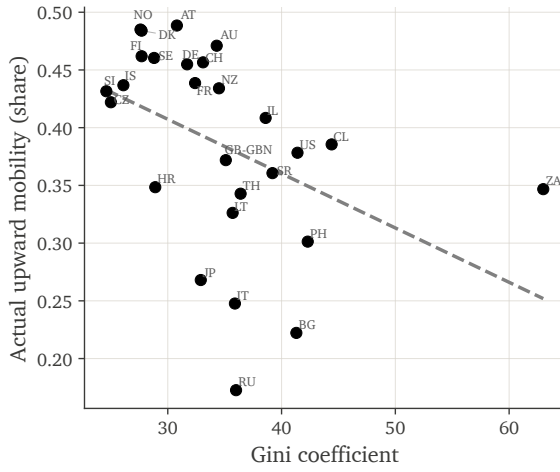
B Robustness of the cross-country belief reversal (Figure 1)

This appendix substantiates the compact robustness claim made in Section 3 for the cross-country belief reversal of Figure 1. The reversal holds under alternative measures of inequality (the top-5% and top-1% income shares in place of the Gini coefficient), under alternative measures of mobility (a continuous status gap, status persistence as in Corak (2013), and a rags-to-riches transition rate), under exclusion of South Africa and the Philippines, the two countries with the highest visual leverage, and under country-level controls for average income, growth, and institutions. We close with the country-level overoptimism gap, the level counterpart to the slope reversal. Each check pairs an actual-mobility panel with an expected-mobility panel measured on the same ladder, so that the opposing signs can be read directly.

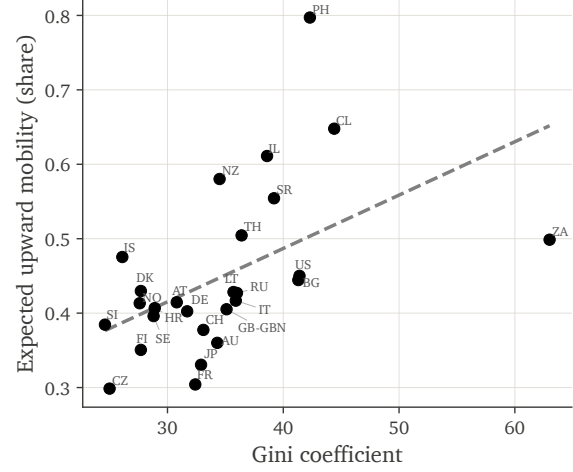
Outlier exclusion: the full 27-country sample

The reversal does not depend on excluding the two highest-leverage countries. Figure B.1 plots the Gini coefficient against the binary mobility shares for the full sample of 27 countries (top row) and after excluding South Africa and the Philippines (bottom row). In every panel the sign is unchanged: actual mobility is lower in more unequal countries, while expected mobility is higher.

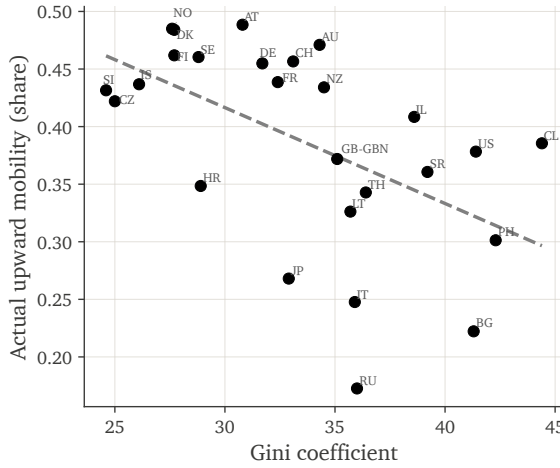
Figure B.1: Gini and binary mobility: full sample and outlier exclusion



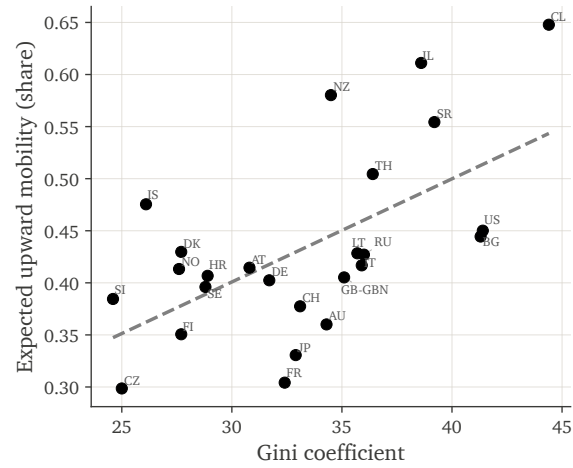
(a) Actual mobility, all 27 countries



(b) Expected mobility, all 27 countries



(c) Actual mobility, outliers excluded



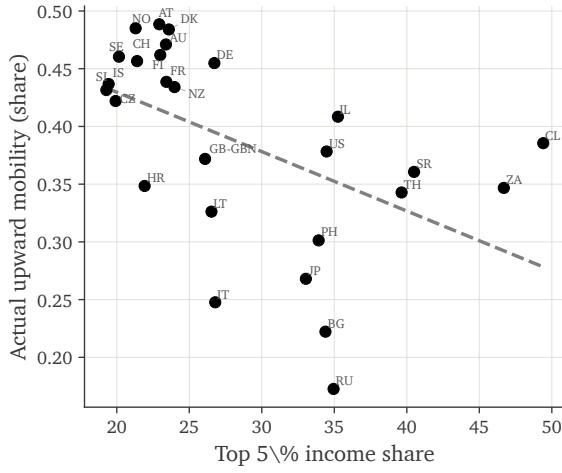
(d) Expected mobility, outliers excluded

Notes: Each point is one of the 27 countries in the 2019 International Social Survey Programme (ISSP), a survey that asks respondents to place their parents, themselves today, and themselves in 10 years on a social status ladder from 1 to 10. Inequality is the Gini coefficient of disposable income. Actual mobility is the country share of respondents who place themselves above their parents ($v_{41} > v_{42}$); expected mobility is the country share who expect a higher status in 10 years than today ($v_{43} > v_{41}$). The top row retains all 27 countries; the bottom row excludes South Africa and the Philippines, the two countries with the highest visual leverage. The dashed line in each panel is the ordinary least squares fit. In every panel actual mobility is lower in more unequal countries and expected mobility is higher, so the belief reversal does not depend on the outlier exclusion.

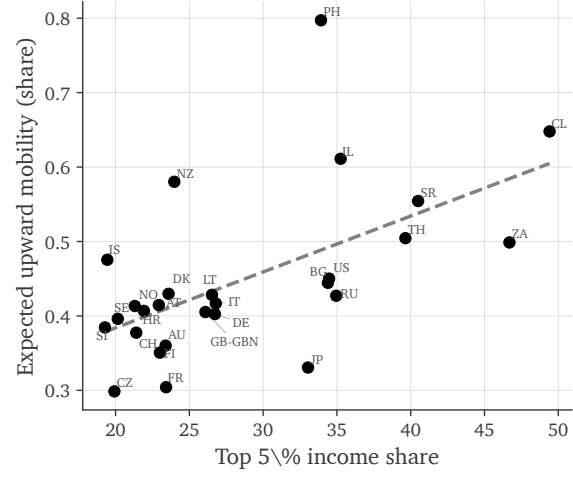
Alternative inequality measures: top-5% and top-1% income shares

The reversal is not specific to the Gini coefficient. Figure B.2 replaces the Gini with the top-5% income share (top row) and the top-1% income share (bottom row). Both signs persist under each concentration measure: actual mobility is lower, and expected mobility is higher, where the income share held at the top is larger.

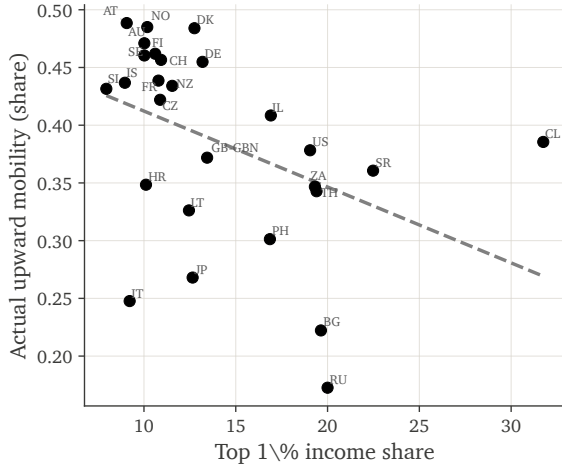
Figure B.2: Top income shares and binary mobility



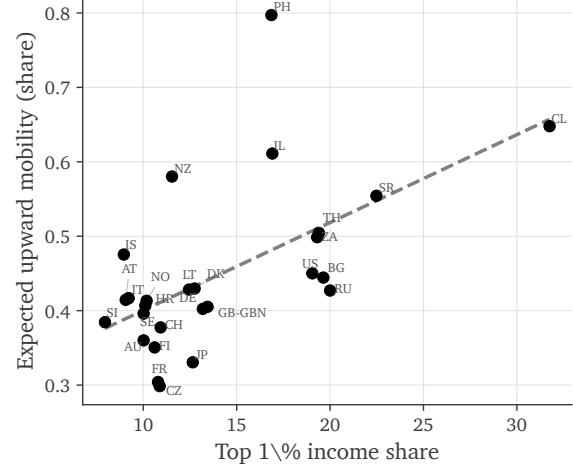
(a) Actual mobility, top-5% share



(b) Expected mobility, top-5% share



(c) Actual mobility, top-1% share



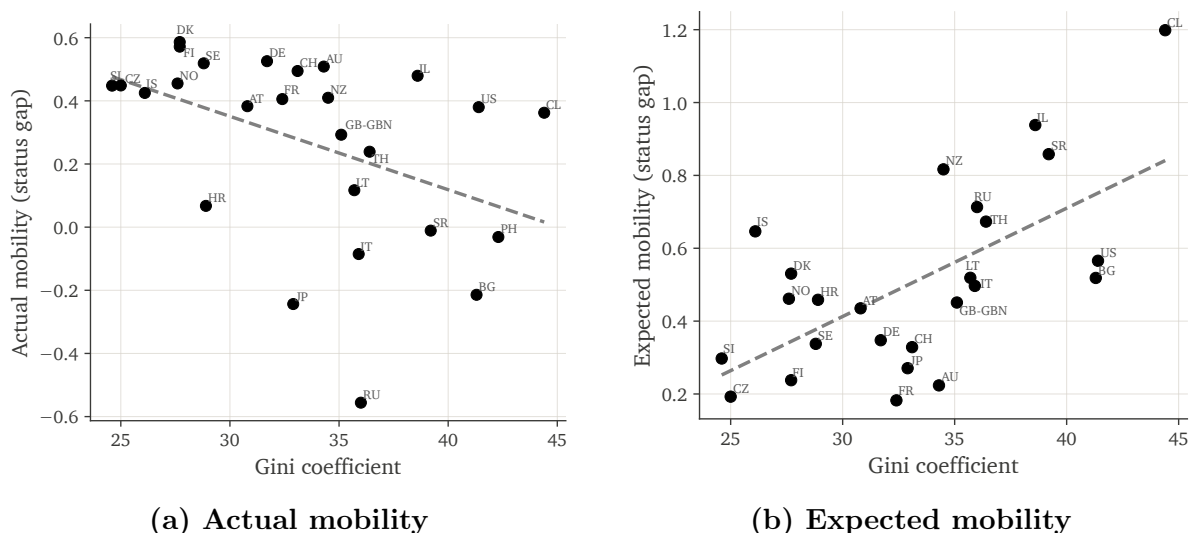
(d) Expected mobility, top-1% share

Notes: Each point is one of the 27 ISSP 2019 countries. Inequality is measured by the top-5% income share (top row) and the top-1% income share (bottom row) from the World Inequality Database (WID), the fraction of national income accruing to the highest 5% and the highest 1% of earners respectively. Actual mobility is the country share of respondents who place themselves above their parents ($v_{41} > v_{42}$); expected mobility is the country share who expect a higher status in 10 years than today ($v_{43} > v_{41}$). The dashed line in each panel is the ordinary least squares fit. Under each top-share measure, actual mobility is lower and expected mobility is higher where inequality is greater.

Alternative mobility measures

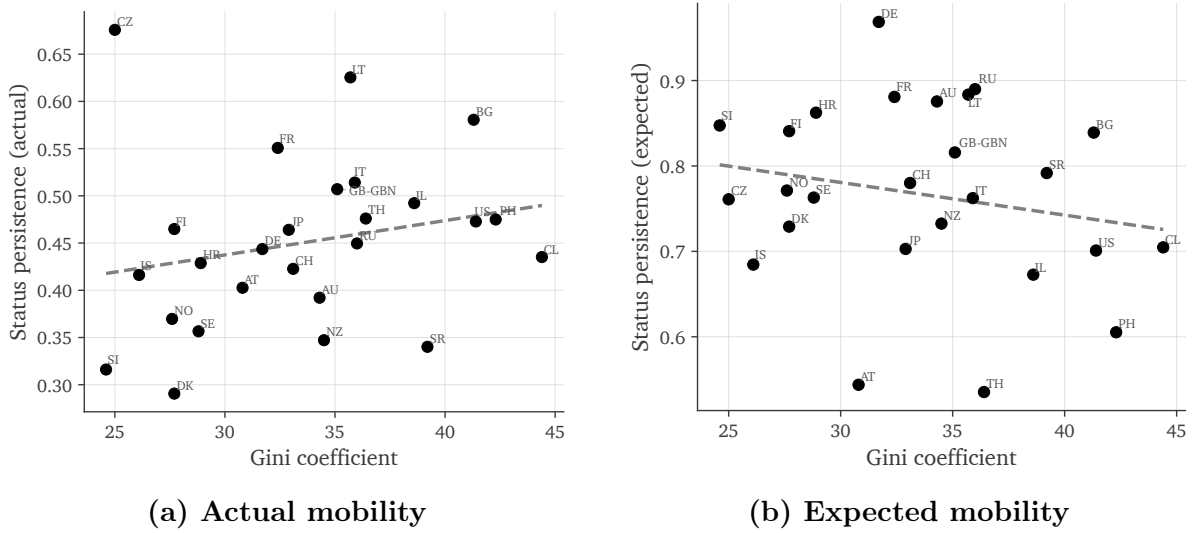
The reversal also holds under alternative measures of mobility. Figure B.3 replaces the binary mobility shares with the continuous status gap, the country mean of $v_{41} - v_{42}$ for actual mobility and of $v_{43} - v_{41}$ for expected mobility. Figure B.4 measures mobility by status persistence in the sense of Corak (2013), the within-country slope of current status on family-of-origin status, where a steeper slope signals less mobility. Figure B.5 measures it by a rags-to-riches transition rate, the share of respondents who move from the bottom of the ladder to the top. In each case the actual measure is associated with lower mobility in more unequal countries while the expected measure is associated with higher mobility.

Figure B.3: Gini and continuous mobility



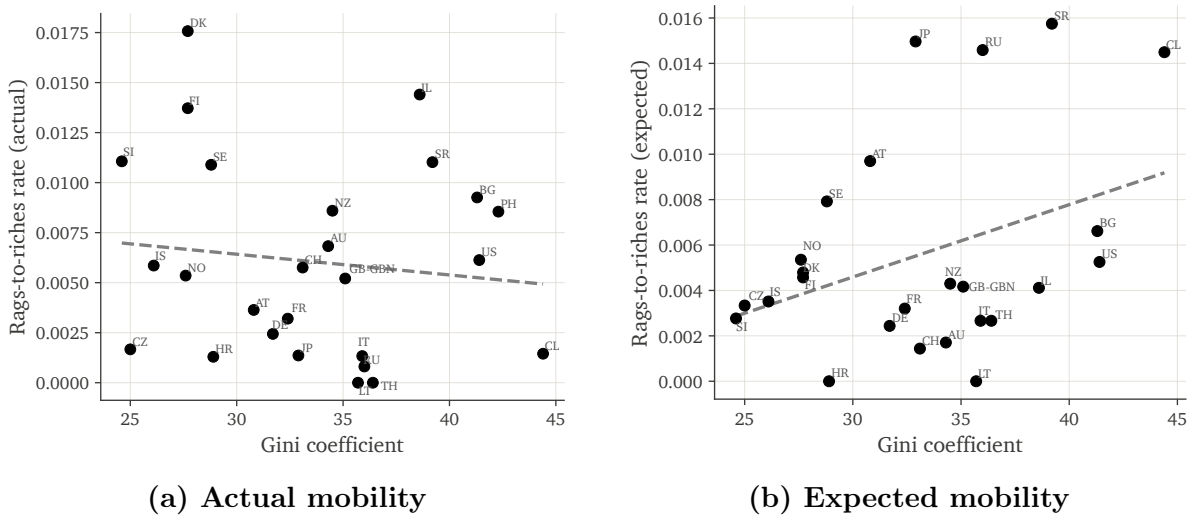
Notes: Each point is an ISSP 2019 country, excluding South Africa and the Philippines, the two countries with the highest visual leverage. Inequality is the Gini coefficient of disposable income. Mobility is measured as a continuous status gap on the 1-to-10 ladder: actual mobility is the country mean of $v_{41} - v_{42}$ (current status minus family-of-origin status); expected mobility is the country mean of $v_{43} - v_{41}$ (expected status in 10 years minus current status). The dashed line in each panel is the ordinary least squares fit. Actual mobility is lower in more unequal countries and expected mobility is higher.

Figure B.4: Gini and status persistence



Notes: Each point is an ISSP 2019 country, excluding South Africa and the Philippines, the two countries with the highest visual leverage. Inequality is the Gini coefficient of disposable income. Mobility is measured by status persistence in the sense of Corak (2013): the within-country slope from a regression of next status on prior status, where a larger slope indicates that status carries over more strongly across the ladder and hence less mobility. The actual panel uses the slope of current status (v_{41}) on family-of-origin status (v_{42}); the expected panel uses the slope of expected status (v_{43}) on current status (v_{41}). The dashed line in each panel is the ordinary least squares fit. Actual persistence is higher in more unequal countries (less actual mobility), while expected persistence is lower (more expected mobility).

Figure B.5: Gini and rags-to-riches transitions

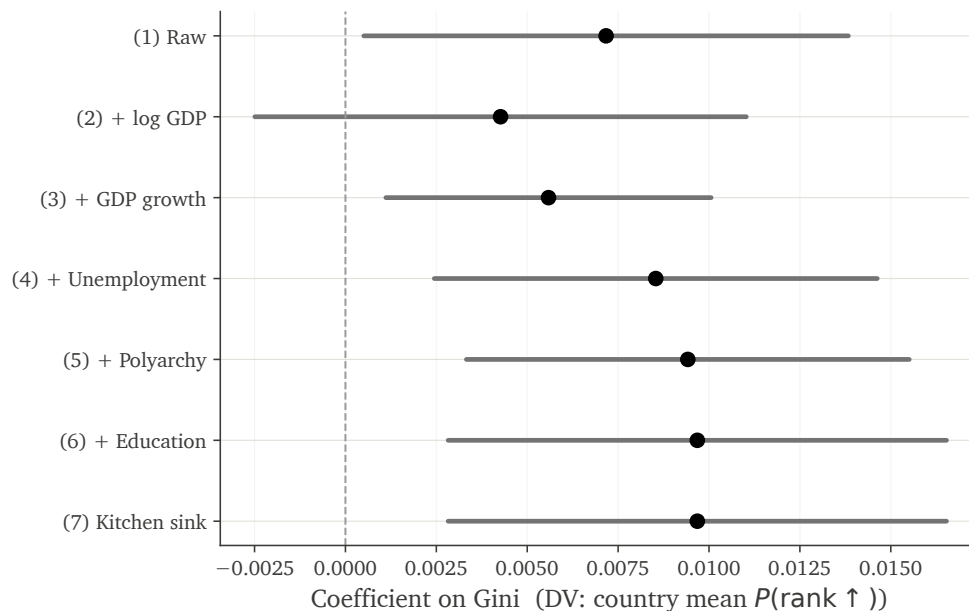


Notes: Each point is an ISSP 2019 country, excluding South Africa and the Philippines, the two countries with the highest visual leverage. Inequality is the Gini coefficient of disposable income. Mobility is measured by a rags-to-riches transition rate, the country share of respondents who move from the bottom of the 1-to-10 ladder to the top, in the spirit of the bottom-to-top transition probabilities of Chetty and coauthors. The actual panel uses transitions from family-of-origin status (v_{42}) to current status (v_{41}); the expected panel uses transitions from current status (v_{41}) to expected status (v_{43}). The dashed line in each panel is the ordinary least squares fit. Actual transitions are less frequent in more unequal countries while expected transitions are more frequent.

Country-level controls: the reversal strengthens

The cross-country reversal is not an artifact of omitted country characteristics. Figure B.6 and Table B.1 regress the country-mean probability of expecting to move up on the Gini coefficient, adding controls in sequence. The coefficient on the Gini is 0.007 ($p < 0.05$) in the raw bivariate regression and 0.010 ($p < 0.01$) under the full control set of log GDP per capita, 10-year GDP growth, unemployment, V-Dem polyarchy, and years of education; the R^2 is 0.26 in the bivariate regression and 0.58 under the full set, with $N = 27$ countries (26 once controls requiring complete data are included). The positive association between inequality and expected mobility is therefore stronger, not weaker, once country characteristics are held constant.

Figure B.6: Gini coefficient on expected upward mobility across control sets



Notes: The dependent variable is the country-mean probability that a respondent expects to move up in social rank over the next 10 years, across the 27 ISSP 2019 countries (including South Africa and the Philippines). Each marker is the estimated coefficient on the Gini coefficient of disposable income from a country-level ordinary least squares regression, with horizontal bars showing 95% confidence intervals based on heteroskedasticity-robust (HC1) standard errors. The specifications add controls progressively, from the raw bivariate regression to the full set of log GDP per capita, 10-year average GDP growth, unemployment, V-Dem polyarchy, and country-mean years of education. The coefficient on the Gini is positive in every specification and larger in magnitude under the fuller control sets; the full table is reported in Table B.1.

Table B.1: Country-level Gini effect on expected upward mobility: kitchen-sink controls.

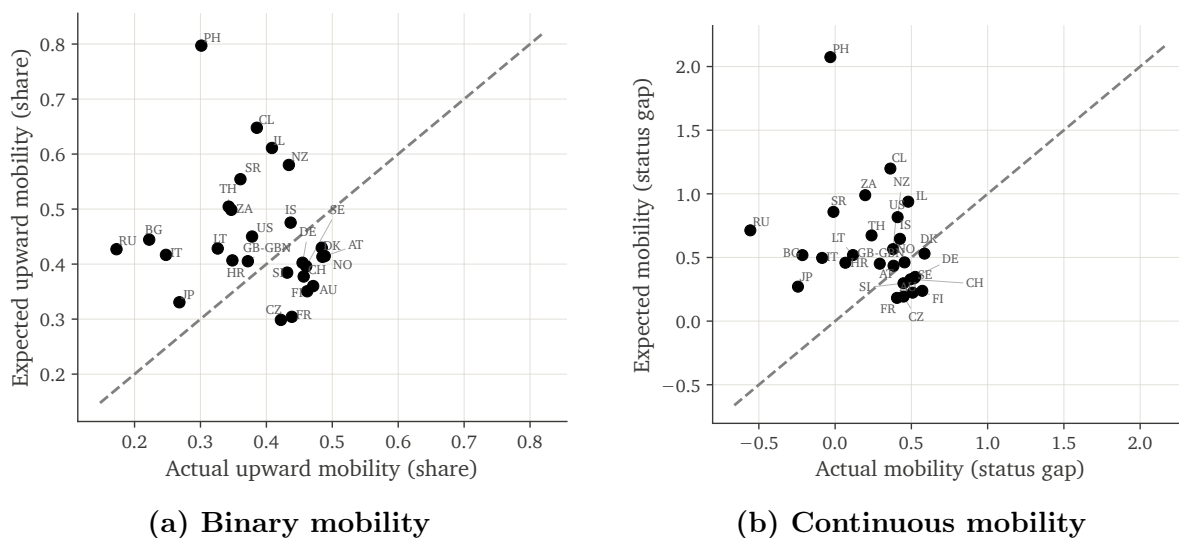
	Dependent variable: country-mean $P(\text{rank } \uparrow)$ (<code>m_moveup_exp</code>)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Gini	0.007** (0.003)	0.004 (0.003)	0.006** (0.002)	0.009*** (0.003)	0.009*** (0.003)	0.010*** (0.003)	0.010*** (0.003)
log GDP per capita		-0.046 (0.032)	0.009 (0.025)	-0.013 (0.033)	-0.051 (0.053)	-0.060 (0.075)	-0.060 (0.075)
GDP growth (10y avg)			6.272** (2.631)	4.127 (2.664)	3.512 (2.735)	3.369 (2.999)	3.369 (2.999)
Unemployment				-0.009 (0.006)	-0.013* (0.007)	-0.014 (0.009)	-0.014 (0.009)
Polyarchy					0.186 (0.146)	0.193 (0.160)	0.193 (0.160)
Education (yrs)						0.004 (0.019)	0.004 (0.019)
Observations	27	27	26	26	26	26	26
R^2	0.258	0.349	0.497	0.547	0.576	0.577	0.577

Notes: Country-level OLS, $N = 27$ ISSP countries (including South Africa and the Philippines). The dependent variable is the country-mean probability that a respondent expects to move up in social rank over the next 10 years (`m_moveup_exp`). Column (1) is the raw bivariate regression. Columns (2)–(6) sequentially add log GDP per capita (Penn World Table), 10-year average GDP growth (Maddison), unemployment, V-Dem polyarchy (2010–2019 mean), and country-mean years of education in the ISSP sample. Column (7) is the kitchen-sink specification (same as col. 6 because all controls are already in). Heteroskedasticity-robust (HC1) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The overoptimism gap

Expected mobility exceeds actual mobility at the country level, and the excess is larger in more unequal countries. Figure B.7 plots each country's expected mobility against its actual mobility, on the binary measure (Panel a) and the continuous measure (Panel b); points above the 45-degree line are countries that expect more mobility than they have realized. Almost every country lies above the line: the mean gap is 0.063 on the binary measure and 0.318 on the continuous measure, and the gap is larger where inequality is higher (correlation with the Gini coefficient of 0.61 on the binary measure and 0.59 on the continuous measure, both $p < 0.01$). The most overoptimistic countries are the Philippines, Chile, and Russia; the least are France, Czechia, and Finland. This level gap is the counterpart to the slope reversal in Figure 1: in more unequal countries beliefs run further ahead of realized mobility. The full country ranking and correlations appear in Table B.2.

Figure B.7: Overoptimism: expected versus actual mobility by country



Notes: Each point is one of the 27 ISSP 2019 countries (including South Africa and the Philippines), plotting expected mobility (vertical axis) against actual mobility (horizontal axis). The dashed line is the 45-degree line; a country above it expects more upward mobility over the next 10 years than its respondents have realized relative to their parents. Panel (a) uses the binary measure: the country share with $v_{41} > v_{42}$ (actual) and with $v_{43} > v_{41}$ (expected). Panel (b) uses the continuous measure: the country mean of $v_{41} - v_{42}$ (actual) and of $v_{43} - v_{41}$ (expected). The mean expected-minus-actual gap is 0.063 on the binary measure and 0.318 on the continuous measure, and the gap is larger in more unequal countries (correlation with the Gini coefficient of 0.61 on the binary measure and 0.59 on the continuous measure, both $p < 0.01$).

Table B.2: Country-level overoptimism: expected minus actual mobility, ranked.

Country	Gini	<i>Binary mobility</i>			<i>Continuous mobility</i>		
		Actual	Expected	Gap	Actual	Expected	Gap
PHL	42.3	0.301	0.797	+0.496	-0.031	+2.074	+2.105
CHL	44.4	0.386	0.648	+0.262	+0.362	+1.199	+0.836
RUS	36.0	0.173	0.427	+0.254	-0.556	+0.713	+1.269
BGR	41.3	0.222	0.444	+0.222	-0.214	+0.519	+0.733
ISR	38.6	0.408	0.611	+0.203	+0.479	+0.938	+0.459
SUR	39.2	0.361	0.554	+0.194	-0.011	+0.858	+0.869
ITA	35.9	0.248	0.417	+0.169	-0.085	+0.497	+0.582
THA	36.4	0.343	0.504	+0.162	+0.239	+0.673	+0.434
ZAF	63.0	0.347	0.499	+0.152	+0.197	+0.989	+0.792
NZL	34.5	0.434	0.580	+0.146	+0.410	+0.817	+0.407
LTU	35.7	0.326	0.428	+0.102	+0.117	+0.519	+0.402
USA	41.4	0.378	0.450	+0.072	+0.380	+0.566	+0.186
JPN	32.9	0.268	0.331	+0.063	-0.244	+0.271	+0.514
HRV	28.9	0.348	0.407	+0.058	+0.067	+0.459	+0.391
ISL	26.1	0.437	0.475	+0.039	+0.425	+0.646	+0.221
GBR	35.1	0.372	0.405	+0.033	+0.293	+0.451	+0.158
SVN	24.6	0.432	0.385	-0.047	+0.448	+0.297	-0.151
DEU	31.7	0.455	0.402	-0.052	+0.526	+0.348	-0.178
DNK	27.7	0.484	0.430	-0.054	+0.586	+0.530	-0.056
SWE	28.8	0.460	0.396	-0.064	+0.519	+0.338	-0.181
NOR	27.6	0.485	0.413	-0.072	+0.455	+0.461	+0.006
AUT	30.8	0.488	0.415	-0.074	+0.383	+0.435	+0.052
CHE	33.1	0.457	0.377	-0.079	+0.494	+0.329	-0.166
AUS	34.3	0.471	0.360	-0.111	+0.509	+0.224	-0.285
FIN	27.7	0.462	0.351	-0.111	+0.572	+0.238	-0.334
CZE	25.0	0.422	0.299	-0.123	+0.449	+0.193	-0.256
FRA	32.4	0.439	0.304	-0.134	+0.406	+0.182	-0.223
Mean	—	0.385	0.449	+0.063	+0.266	+0.584	+0.318

Notes: 27 ISSP 2019 countries (including ZA and PH), sorted from most to least overoptimistic on the binary measure. *Binary mobility*: country mean of $\mathbf{1}\{v_{41} > v_{42}\}$ (actual) and $\mathbf{1}\{v_{43} > v_{41}\}$ (expected). *Continuous mobility*: country mean of $v_{41} - v_{42}$ (actual) and $v_{43} - v_{41}$ (expected). Gap = expected minus actual; a positive value indicates the country's respondents expect more upward mobility over the next 10 years than they have experienced from their parents. Pearson correlations of the gap with country inequality: $\text{corr}(\text{binary gap, Gini}) = +0.61$; $\text{corr}(\text{binary gap, top-5\%}) = +0.70$; $\text{corr}(\text{continuous gap, Gini}) = +0.59$; $\text{corr}(\text{continuous gap, top-5\%}) = +0.65$. All four are significant at $p < 0.01$ (one-tailed).

C Additional figures and tables - empirical evidence (ISSP survey data)

Table C.1: Descriptive statistics for main variables of interest in ISSP dataset

	Countries with below median Gini					Countries with above median Gini				
	Mean	SD	Min	Max	N	Mean	SD	Min	Max	N
Status: current	5.73	1.68	1.00	10.00	12433	5.09	1.86	1.00	10.00	11628
Status: Family's when growing up	5.30	1.78	1.00	10.00	12433	4.93	1.99	1.00	10.00	11628
Status: Expected in 10 years	6.03	1.90	1.00	10.00	12433	6.02	2.11	1.00	10.00	11628
Rich should pay higher taxes than poor	3.96	0.76	1.00	5.00	12433	3.95	0.86	1.00	5.00	11628
Age	46.19	10.83	26.00	65.00	12433	44.73	10.89	26.00	65.00	11628
Male	0.48	0.50	0.00	1.00	12415	0.46	0.50	0.00	1.00	11621
Years of Education	14.58	4.02	0.00	57.00	12228	11.87	4.31	0.00	52.00	11237
Experienced inequality	0.21	0.05	0.10	0.39	12433	0.34	0.09	0.15	0.52	11628
Experienced unemployment	5.90	2.98	0.37	14.53	11152	8.52	5.92	0.87	26.75	10509

Countries with below median Gini: Australia, Austria, Republic of Croatia, Czech Republic, Denmark, Finland, France, Germany, Iceland, New Zealand, Norway, Russian Federation, Republic of Slovenia, Sweden, Switzerland; Countries with above median Gini: Bulgaria, Chile, Israel, Italy, Japan, Republic of Lithuania, Philippines, South Africa, Suriname, Thailand, United Kingdom, United States; Experienced inequality and experienced unemployment are constructed as in Roth and Wohlfahrt (2018).

C.1 Experienced inequality: full results and robustness

This appendix collects the supporting material for the experienced-inequality analysis of Section 3.2. Table C.2 reports the four-specification estimates of the effect of experienced inequality on expected mobility that underlie Figure 2, and Table C.3 reports the coefficient on every non-fixed-effect regressor in those specifications. Every coefficient discussed below reproduces across Python and Stata (`reghdfe`) to four decimal places.

Functional form. Figure C.1 residualizes both the experienced-inequality regressor and expected mobility on the full set of controls and fixed effects and bins the residuals. The relationship is linear throughout, with no kink, threshold, or curvature, so the linear specification is appropriate.

Where the mass moves: per-rung decomposition. Figure C.2 estimates the effect of experienced inequality on the probability of expecting each destination rung separately. The positive loadings concentrate at the top of the ladder (rungs 8 to 10), with negative loadings in the upper-middle (the at-mean rung 7) and at the bottom, rather than a uniform shift.

Mapping rungs to income, and the above-mean test. Figure C.3 maps each ladder rung to the average within-country household income of respondents at that rung, locating the country-mean income between rungs 7 and 8. Figure C.4 then shows the sharp prediction: experienced inequality raises the probability of expecting an above-mean-income destination and lowers the probability of an at-or-below-mean destination by an equal amount.

Heterogeneity. The effect is positive across age cohorts and across household-income quintiles, and it is concentrated among lower-status respondents, where the prospect of climbing

Table C.2: Effect of experienced inequality on expected mobility

	<i>Dependent variable: Expected mobility</i>			
	(1)	(2)	(3)	(4)
Experienced inequality	0.466*** (0.135)	0.335*** (0.113)	0.288*** (0.104)	0.252** (0.111)
Individual controls		Yes	Yes	Yes
Country FE			Yes	Yes
Subregion×cohort FE			Yes	Yes
All-experience controls				Yes
<i>N</i>	22,558	21,879	21,879	19,152

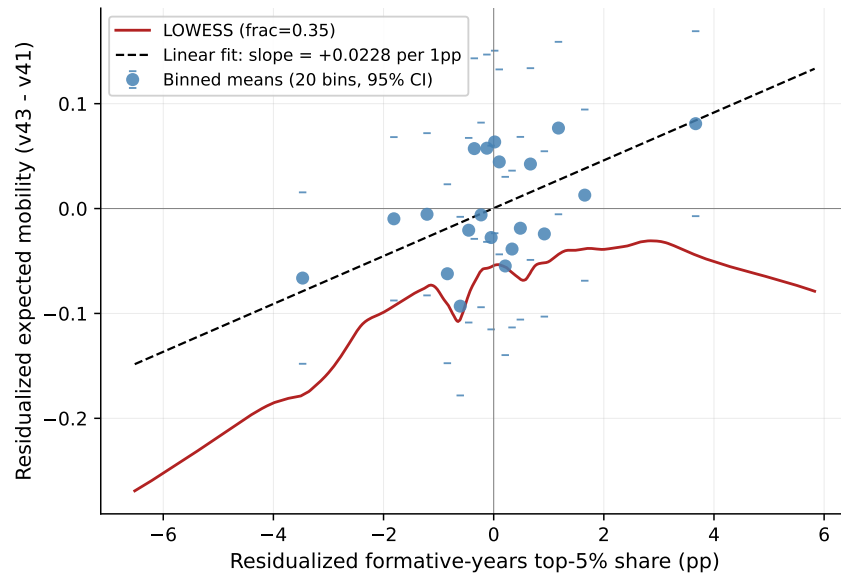
Notes: The dependent variable is expected mobility, defined as the respondent’s expected social status ten years from now minus their current self-reported social status (each on a 1–10 ladder). The regressor of interest is the average top-5% income share in the respondent’s country over ages 18–25 (Roth–Wohlfahrt formative-years measure), standardized within sample. The top-5% share captures how much more, in income, the top 5% earn relative to the population mean. Sample: ISSP 2019 respondents aged 29 or older. Individual controls: age, female, and years of education. “All-experience controls” add same-window averages of country-level lifetime experiences (unemployment, GDP per capita growth, CPI inflation, electoral democracy, banking-or-currency-crisis exposure, rare-disaster exposure) and family-of-origin past status. The appendix table reports the coefficients on each control. Standard errors clustered at the country level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

is most relevant, turning negative only in the small top-status bin (Figure C.5). It also holds across alternative inequality measures, with the top-1% and top-10% shares giving the same pattern as the top-5% (Figure C.6). Adding the six lifetime-experience controls leaves each of these patterns unchanged.

Placebo windows. Figure C.7 re-estimates the headline using inequality experienced in windows other than the formative years. Inequality experienced before the formative window (ages 0 to 7) and after it (ages 34 to 41) carries no effect; only the 18 to 25 window does.

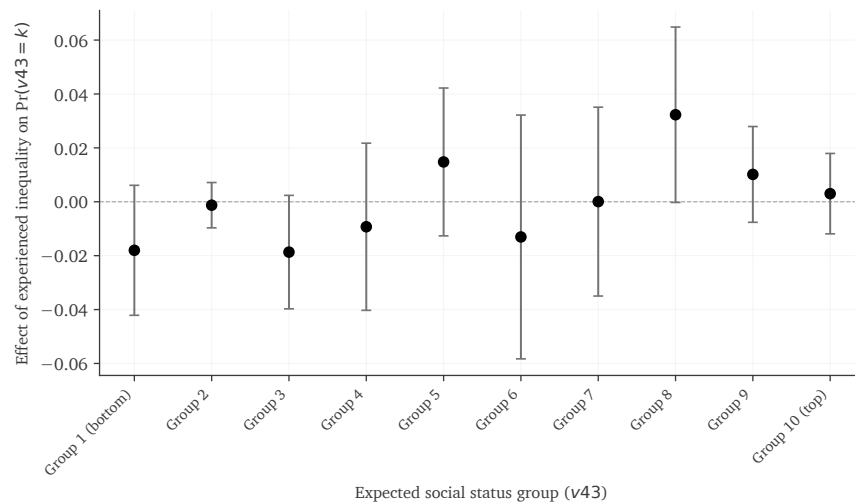
C.2 Additional survey evidence: status distributions and the experienced-inequality measure

Figure C.1: Residualized binscatter of expected mobility on experienced inequality



Notes: Both variables are residualized on country fixed effects, subregion-by-cohort fixed effects, age, female, and education. Blue points are means within 20 equal-sized bins of the residualized regressor; the red curve is a LOWESS smoother; the dashed line is the OLS fit on the residuals.

Figure C.2: Effect of experienced inequality on the probability of each expected destination rung



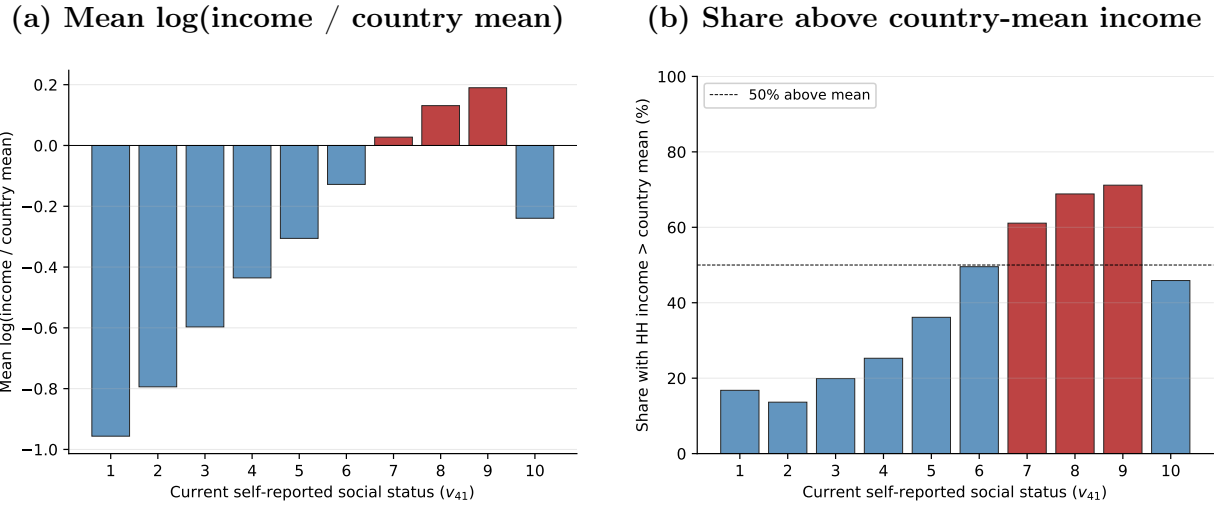
Notes: Each point is the coefficient on standardized experienced inequality from a separate linear-probability regression of an indicator for expecting destination rung k , with the headline controls and fixed effects. 95% confidence intervals clustered at the country level.

Table C.3: Effect of experienced inequality on expected mobility — full coefficients

	<i>Dependent variable: Expected mobility</i>			
	(1)	(2)	(3)	(4)
Experienced inequality	0.466*** (0.135)	0.335*** (0.113)	0.288*** (0.104)	0.252** (0.111)
Age	—	-0.035*** (0.003)	-0.047*** (0.007)	-0.044*** (0.007)
Female	—	-0.014 (0.019)	-0.002 (0.024)	0.002 (0.023)
Years of education	—	-0.033*** (0.012)	-0.012** (0.005)	-0.005 (0.006)
Family-of-origin past status	—	—	—	-0.044*** (0.009)
Experienced unemployment rate	—	—	—	0.049 (0.036)
Experienced GDP per capita growth	—	—	—	0.046* (0.024)
Experienced CPI inflation	—	—	—	-0.014 (0.023)
Experienced electoral democracy	—	—	—	-0.079 (0.048)
Experienced banking/currency crisis	—	—	—	-0.005 (0.014)
Experienced rare disaster	—	—	—	0.015 (0.017)
Country FE			Yes	Yes
Subregion×cohort FE			Yes	Yes
<i>N</i>	22,558	21,879	21,879	19,152

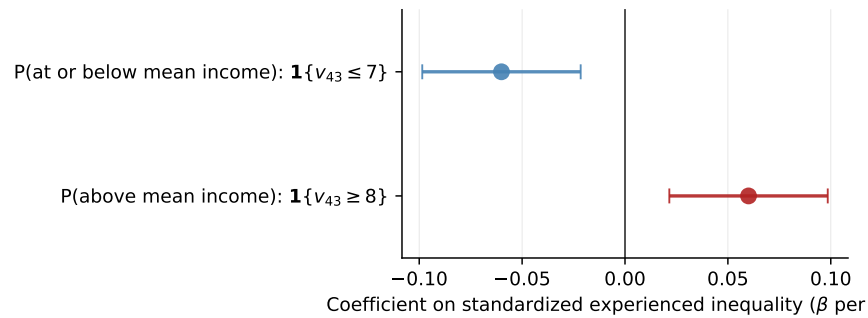
Notes: Companion to Table C.2 reporting the coefficient and standard error on every non-fixed-effect variable in each specification. The dependent variable is expected mobility, defined as the respondent’s expected social status ten years from now minus their current self-reported social status (each on a 1–10 ladder). The independent variable of interest, “Experienced inequality”, is the average top-5% income share in the respondent’s country over ages 18–25, standardized within sample. The “Experienced” macro variables in column (4) are constructed identically (same-window averages, then standardized). Sample: ISSP 2019 respondents aged 29 or older. Standard errors clustered at the country level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure C.3: Mapping current status to within-country household income



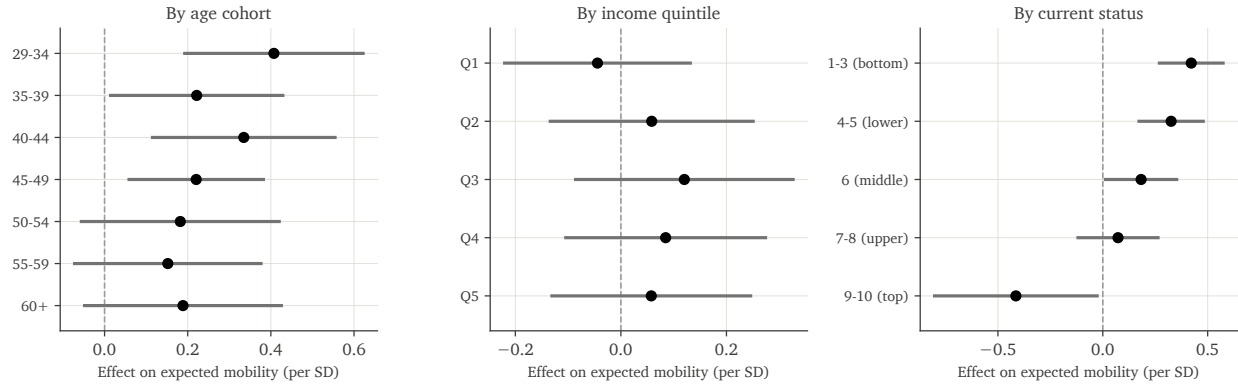
Notes: Within-country household income position at each rung of current status v_{41} . Panel (a): mean of log(income/country mean) within each rung; zero marks the country mean. Panel (b): share at each rung whose household income exceeds the country mean. Rungs 1 to 7 sit at or below the country mean; rungs 8 and above sit clearly above it.

Figure C.4: Effect on the probability of an above-mean versus at-or-below-mean expected destination



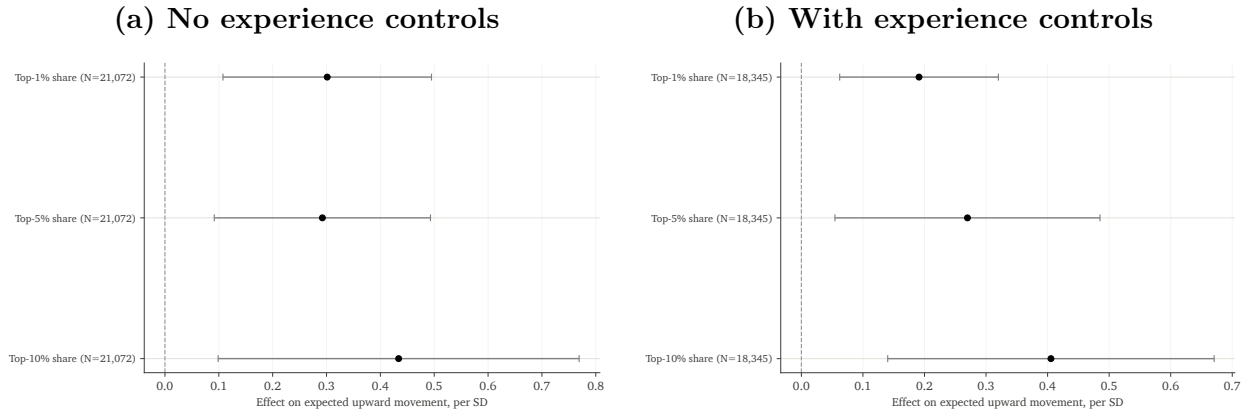
Notes: Coefficient on standardized experienced inequality from separate linear-probability models for an above-country-mean expected destination ($v_{43} \geq 8$) and an at-or-below-mean destination ($v_{43} \leq 7$). The two are equal and opposite by construction. Controls and fixed effects as in the headline specification; 95% confidence intervals clustered at the country level.

Figure C.5: Experienced inequality and expected mobility: heterogeneity by age, income, and status



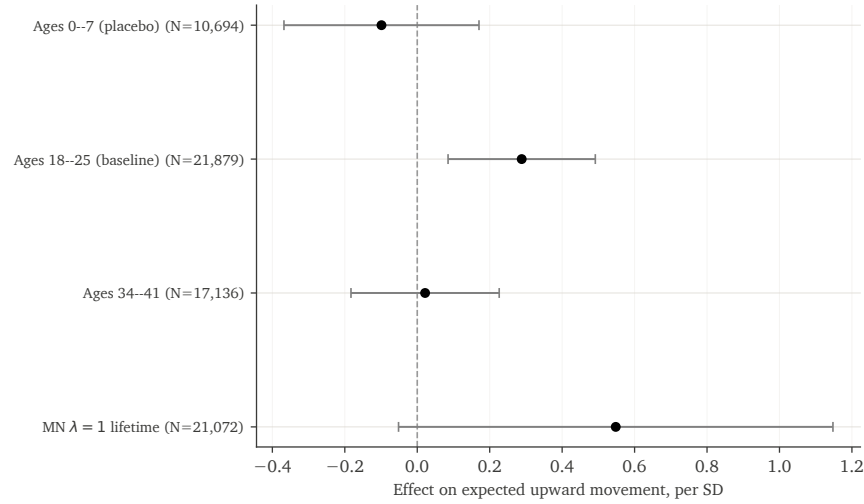
Notes: Within-bin coefficient on standardized experienced inequality (per standard deviation) on expected mobility ($v_{43} - v_{41}$), from a full-sample interaction model that lets the slope vary across the bins of each dimension while preserving the headline country and subregion-by-cohort fixed effects and individual controls. Left: seven age cohorts; middle: within-country household-income quintiles; right: current social status (v_{41}). The effect is positive across age cohorts and income quintiles and is concentrated among lower-status respondents, turning negative only in the small top-status bin ($N = 673$). Adding the six lifetime-experience controls leaves the pattern unchanged. 95% confidence intervals clustered at the country level. Sample: ISSP 2019 respondents aged 29 or older.

Figure C.6: Effect by alternative top-share measure



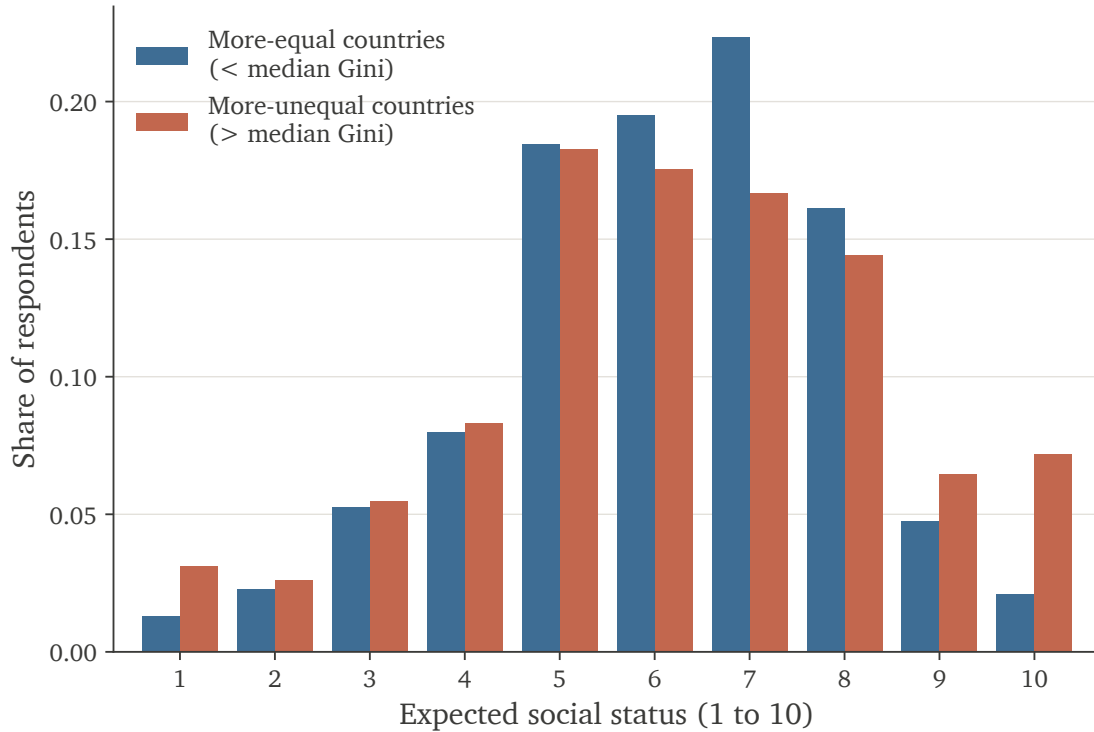
Notes: Coefficient on a standardized formative-years top-1%, top-5%, or top-10% share, each estimated separately on a common sample with the headline controls and fixed effects. 95% confidence intervals clustered at the country level.

Figure C.7: Effect across alternative experience windows



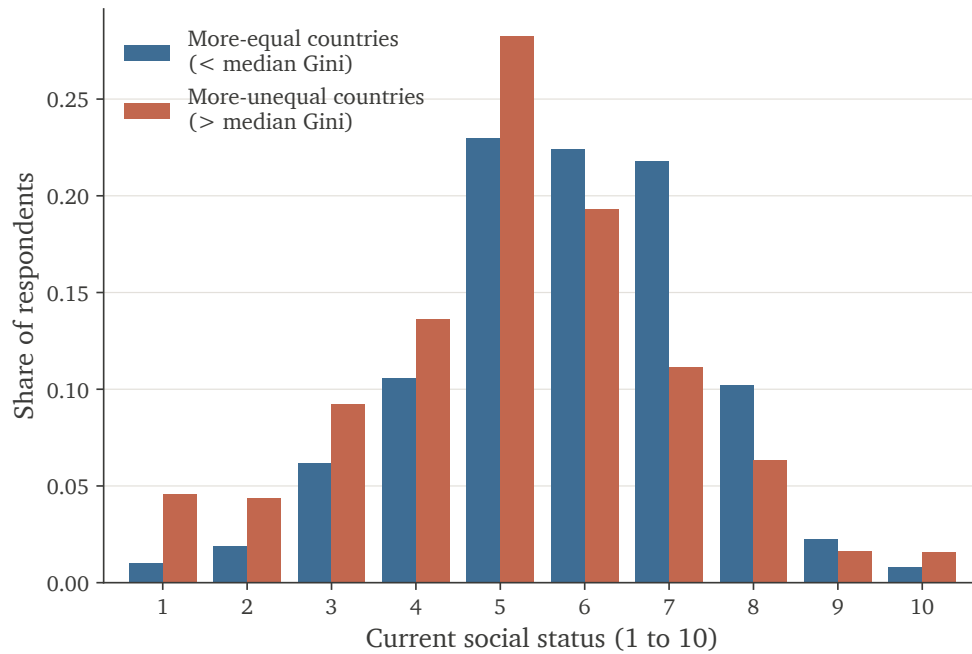
Notes: Each row is a separate regression with the headline controls and fixed effects but a different definition of the experienced-inequality regressor: the baseline ages 18 to 25 window; a pre-formative placebo (ages 0 to 7); a post-formative window (ages 34 to 41); and a Malmendier-Nagel lifetime-weighted average. All regressors are standardized. 95% confidence intervals clustered at the country level.

Figure C.8: Distribution of expected social status, equal versus unequal countries



Notes: Distribution of expected social status in 10 years (1 to 10), for countries below the median Gini (red) and above it (blue). The two distributions diverge at the top of the ladder.

Figure C.9: Distribution of current social status, equal versus unequal countries



Notes: Distribution of current social status (1 to 10), for countries below the median Gini (red) and above it (blue). Unlike expected status, current placement does not diverge at the top.

Table C.4: Logistic regression predicting social status categories by Gini coefficient

	(1) Fut. Bottom (1-6)	(2) Fut. Mid (7-8)	(3) Fut. Top (9-10)	(4) Curr. Bottom (1-5)	(5) Curr. Mid (6-8)	(6) Curr. Top (9-10)
> median Gini	0.062** (0.021)	-0.277*** (0.023)	0.551*** (0.038)	0.603*** (0.021)	-0.603*** (0.021)	-0.047 (0.061)
Constant	0.361*** (0.015)	-0.651*** (0.015)	-2.618*** (0.029)	-0.197*** (0.015)	0.072*** (0.015)	-3.437*** (0.042)
Num.Obs.	36 038	36 038	36 038	36 038	36 038	36 038
Log.Lik.	-24 299.094	-22 352.122	-10 743.910	-24 539.588	-24 375.749	-4917.472
F	8.422	147.027	213.611	796.004	788.258	0.595

Logistic regression estimates predicting probabilities of reporting future and current social status categories (binary). Standard errors reported in parentheses. Above median Gini is a dummy indicating countries with above median income inequality.

C.3 Mobility beliefs and happiness (General Social Survey)

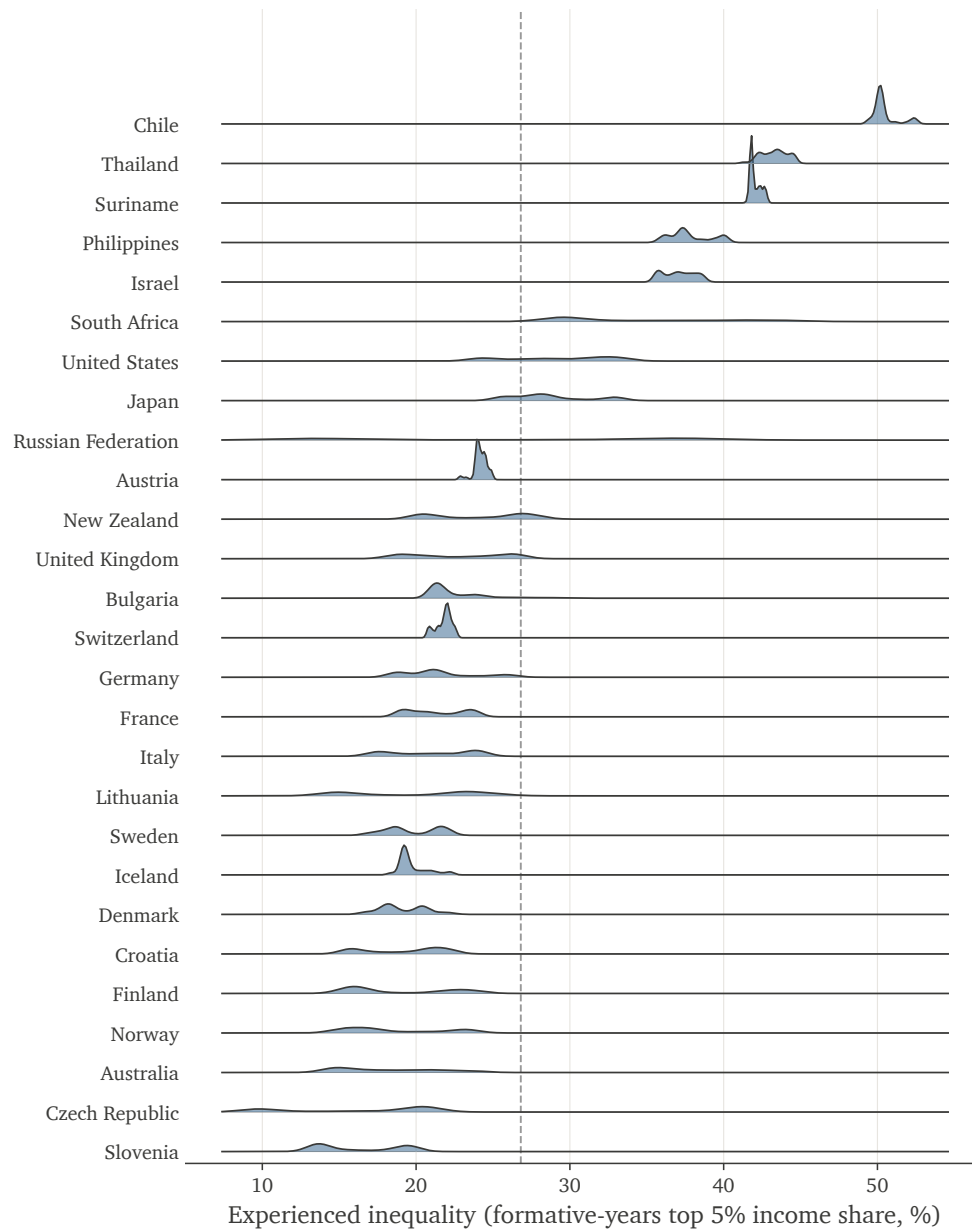
Table C.5: Expected status and general happiness, by class (columns 2-4) and by income (columns 5-7)

	(1) all	(2) lower	(3) middle	(4) upper	(5) low inc.	(6) middle inc.	(7) high inc.
Rank in 10 yrs	0.034** (0.011)	0.053** (0.018)	0.019 (0.019)	0.023 (0.022)	0.054*** (0.016)	0.001 (0.020)	0.026 (0.026)
Rank current	0.020 (0.013)	0.008 (0.030)		0.013 (0.038)	0.006 (0.018)	0.034 (0.024)	0.025 (0.030)
Education	0.002 (0.007)	0.002 (0.013)	0.002 (0.011)	0.001 (0.012)	0.006 (0.011)	-0.015 (0.012)	0.012 (0.013)
Income	0.013*** (0.003)	0.009 (0.006)	0.013* (0.005)	0.018** (0.006)	0.006 (0.006)	0.005 (0.026)	0.020 (0.018)
Gender	0.013 (0.034)	0.043 (0.066)	0.025 (0.053)	-0.024 (0.059)	0.054 (0.061)	0.034 (0.057)	-0.039 (0.057)
Constant	1.304*** (0.120)	1.266*** (0.225)	1.507*** (0.213)	1.414*** (0.349)	1.220*** (0.209)	1.778*** (0.532)	1.093* (0.452)
Num.Obs.	1596	440	629	527	513	540	543
R2	0.041	0.036	0.014	0.027	0.040	0.007	0.021

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

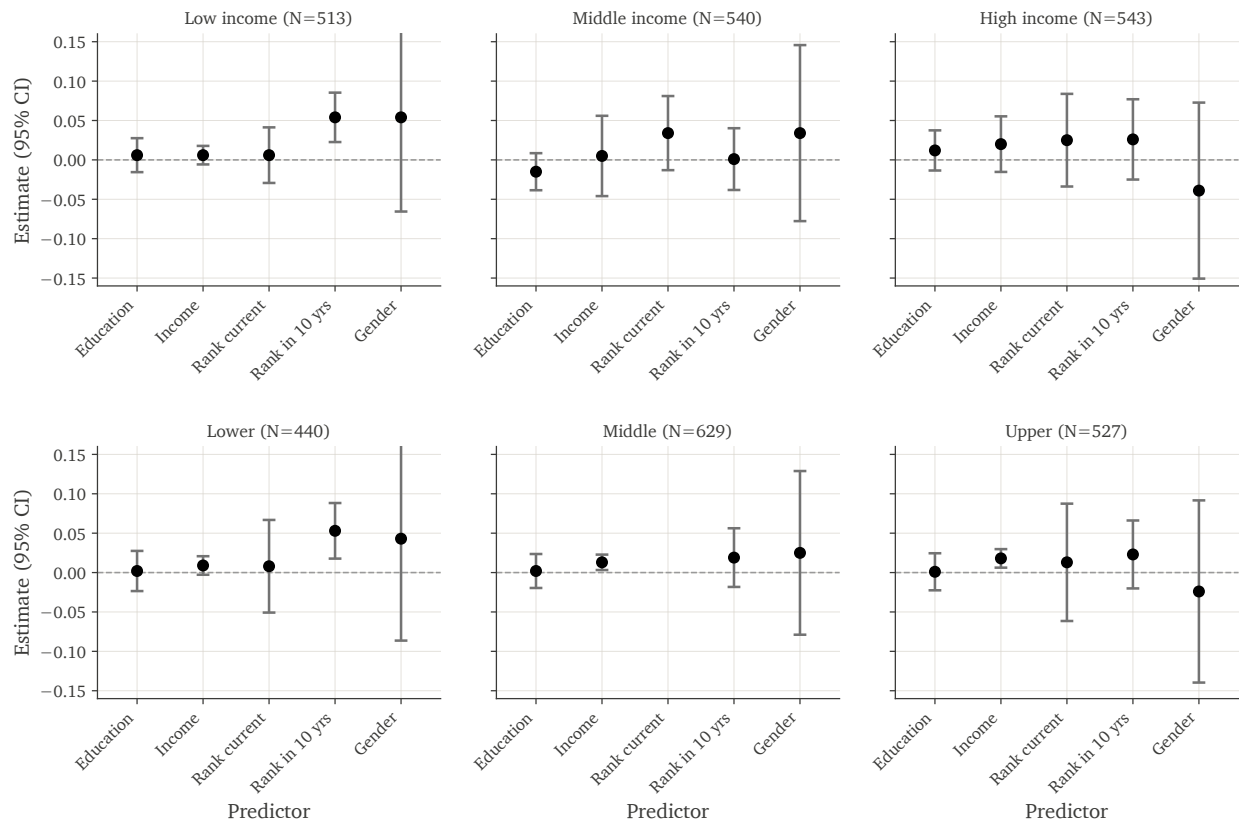
OLS regression estimates of the relationship between expected social status (Rank in 10 years) and general happiness, analyzed by self-placed current rank (columns 2-4) and income groups (columns 5-7). General happiness is measured on a 3-point scale: 1 (not happy at all), 2 (pretty happy), and 3 (very happy). Controls include current rank, education, income, and gender. Standard errors are in parentheses. Sample consists of US citizens from the General Social Survey.

Figure C.10: Distribution of experienced inequality by country



Notes: Distribution of experienced inequality by country, constructed as in Roth and Wohlfart (2018): the top-5% income share in respondent *i*'s country during *i*'s formative years (ages 18 to 25).

Figure C.11: Expected status and general happiness



Notes: Linear regression of general happiness (1 not happy at all, 2 pretty happy, 3 very happy) on expected future status, by income group. US sample, General Social Survey.

D Additional details on the experimental design

Variations of the Baseline treatment. The Baseline treatment has two variations based on when subjects complete the quiz. In our theoretical model, subjects make their effort choice first without knowing about the income allocation choice and take this effort choice as given when choosing the allocation. The first variation of the Baseline treatment, which we refer to as the *Quiz-First* variation, closely follows this model. In this variation, subjects complete two example quiz questions and all 10 main quiz questions *before* the belief elicitation in Period 1 and the income allocation choice in Period 2 (see Figure 4).

In the second variation, which we refer to as the *Quiz-Last* variation, we test whether our results are robust to the timing of the effort choice. Here, subjects first complete two example quiz questions and state their beliefs, then make the Period 2 income allocation choice, and finally complete the 10 quiz questions (see Figure D.1). This allows subjects to jointly determine their preferred inequality and effort level when making the allocation choice.

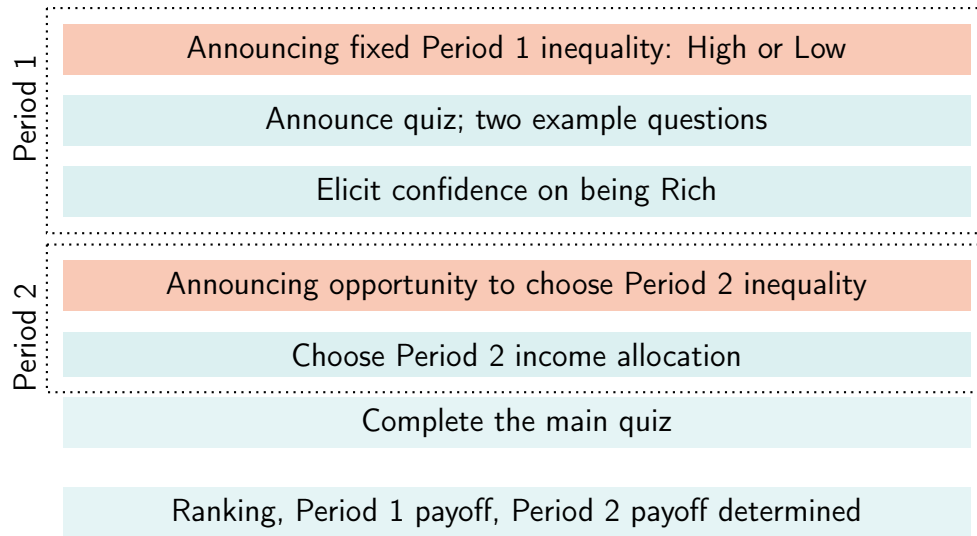


Figure D.1: Experimental Design - Baseline Quiz-Last

In our main analysis, we pool both variations of the Baseline treatment. As a robustness check, we also conduct an analysis comparing only the Quiz-First variation to the Feedback treatment, as these share the same quiz timing.

E Proofs

E.1 Proof of median $P(\text{Rank}2)$ result

To quantify what improved performance would predict, we use the empirical rank probabilities for each quiz score from our data (see Appendix Table A.3). Consider a subject who correctly answers 7 questions (the median performance). Their objective probabilities are $P(\text{Rank } 1) = 5.4\%$ and $P(\text{Rank } 2) = 23.6\%$. If this subject overestimated their performance under high inequality, believing they would answer 8 questions correctly instead of 7, their subjective probabilities would become $P(\text{Rank } 1) = 13.2\%$ and $P(\text{Rank } 2) = 33.8\%$. There is a substantial increase in the probability of achieving each of the top 2 ranks.

Aggregating across all subjects, if everyone believes their performance improves by one correct answer under high inequality while everybody else's performance is constant on average, the average Rank 1 belief would increase by 7.3 percentage points and average Rank 2 belief would increase by 4.5 percentage points. The median Rank 1 belief would increase by 6.3 percentage points (from 8.3% to 14.6%) and median Rank 2 belief would increase by 8.1 percentage points (from 22.9% to 31.0%). Even if subjects believe their performance improves by two correct answers, we reach similar conclusions: average Rank 2 probability should increase by 5.3 percentage points and median Rank 2 probability should increase by 14.3 percentage points (from 22.9% to 37.2%). The prediction of increased Rank 2 beliefs is remarkably robust: the average would remain essentially flat only under the extreme assumption that most subjects believe their performance improves by three or more questions (which would require median performers to believe they scored 10/10), yet even then the median Rank 2 belief would still increase.

We can generalize this observation beyond the specific empirical distribution. Even without assuming subjects know the exact performance distribution, we can prove that, with equilibrium neglect, median Rank 2 beliefs increase under mild assumptions about believed performance distribution:

Assumption 1: $F(s_{med} + t) < 3/4$, where s_{med} is the median score and t is the believed performance improvement.

This condition requires that the median performer, after believed improvement, remains below the 75th percentile of the original distribution.

Given Assumption 1, we can establish the following result: If all individuals believe their own score increases by t units while others' scores remain unchanged, then the median of the distribution of $P(\text{Rank } 2)$ beliefs must increase. The intuition is straightforward: for someone to achieve Rank 2, they need exactly one opponent to outperform them and three to underperform. This probability follows an inverse U-shaped curve as a function of one's position in the distribution: it's low for both very weak performers (who rarely beat three opponents) and very strong performers (who usually beat all four). The probability peaks around the 60th-70th percentile. Since the median performer sits at the 50th percentile, improving their believed performance moves them closer to this peak, increasing their $P(\text{Rank } 2)$.

2). As long as they don't jump all the way past the 75th percentile (Assumption 1), this relationship holds.

Each individual's probability of achieving rank k given score s_i is:

$$P(\text{Rank } k \mid s_i) = C_4^{k-1} [1 - F(s_i)]^{k-1} [F(s_i)]^{5-k}$$

For an individual with score s , the change in $P(\text{Rank } 2)$ when believing their score increases from s to $s + t$ is:

$$\Delta P(\text{Rank } 2 \mid s, t) = 4 \left[[1 - F(s+t)][F(s+t)]^3 - [1 - F(s)][F(s)]^3 \right]$$

Define $g(x) = x^3(1 - x)$. The derivative $g'(x) = 3x^2 - 4x^3 = x^2(3 - 4x)$ is positive for $x < 3/4$ and negative for $x > 3/4$. This means $g(x)$ is strictly increasing on $[0, 3/4]$ and strictly decreasing on $[3/4, 1]$. For the median individual with score s_{med} , we know that $F(s_{med}) \leq F(s_{med} + t) < 3/4$ by Assumption 1.

Since both $F(s_{med})$ and $F(s_{med} + t)$ lie in the region where g is strictly increasing, and since $F(s_{med} + t) \geq F(s_{med})$, we can conclude:

$$g(F(s_{med} + t)) \geq g(F(s_{med}))$$

Therefore $\Delta P(\text{Rank } 2 \mid s_{med}, t) \geq 0$, meaning the median individual's $P(\text{Rank } 2)$ belief weakly increases under the performance improvement hypothesis. The increase becomes strict if we instead assume that $F(s_{med} + t) > F(s_{med})$. Since this individual represents the median of the distribution, the median of all $P(\text{Rank } 2)$ beliefs must increase.

F Conceptual framework with anticipated Period-2 decisions

The conceptual framework in Section 2 assumes subjects form their Period-1 beliefs without anticipating the Period-2 allocation task. This matches the Baseline experimental design, in which Period 2 is introduced only after beliefs are stated. In many real-world settings, however, individuals form beliefs knowing in advance that those beliefs will guide a consequential downstream decision: voters anticipating their electoral choices, investors anticipating their portfolio allocations, students anticipating their occupational entry. This appendix shows that the central predictions of the baseline model, strict overconfidence about Rank 1 and a positive inequality-overconfidence relationship, survive when subjects fully anticipate the Period-2 decision. The motivated-belief mechanism is not an artifact of the Baseline subjects' ignorance of the allocation task; it would arise even under full awareness.

F.1 Setup

We retain all primitives of the baseline model and add a Period-2 allocation task whose consequence subjects fully anticipate. There are n individuals, n ranked positions, one Rich slot ($k = 1$) paying y_R , and $n - 1$ Poor slots each paying y_P . Inequality $\pi \in [1/n, 1]$ is the share of total Period-1 income Y going to Rank 1, so $y_R = \pi Y$, $y_P = (1 - \pi)Y/(n - 1)$, and the Period-1 mean income $\bar{y} = Y/n$ is independent of π . Each individual i has objective probability p_{0ij} of position j with $\sum_j p_{0ij} = 1$.

Period 2. Each subject i chooses an allocation $s_i \in [0, Y_2]$, the dollar amount of a Period-2 budget Y_2 that goes to Rank 1; the remainder $Y_2 - s_i$ is split equally across the $n - 1$ Poor positions. Under random dictator, one of the n subjects is drawn uniformly and her choice is implemented for the entire group. Position 1 receives the implemented allocation $\bar{s}$ and each of positions $2, \dots, n$ receives $(Y_2 - \bar{s})/(n - 1)$.

Payment structure. Final payoff is paid from Period-1 income with probability $1 - \lambda$ and from Period-2 income with probability $\lambda \in (0, 1)$. The two periods are exclusive substitutes: introducing Period 2 reweights the payment lottery rather than adding to it.

Optimal Period-2 allocation. Conditional on subjective belief μ_{i1} about Rank 1, the privately optimal s_i is the Bayesian choice that maximizes expected Period-2 consumption utility under μ_{i1} . Under standard specifications, the optimum is

$$s_i^*(\mu_{i1}) = \mu_{i1} Y_2. \quad (4)$$

Well-being. Subjects choose μ_i to maximize anticipatory utility net of decision cost. The Baseline well-being function is unchanged from Equation (1) of Section 2; the anticipated-decision specification incorporates Period-2 anticipatory utility and uses a decision-cost coefficient $\alpha^A > 0$:

$$W_i^A(\mu_i) = \sum_{j=1}^n \mu_{ij} [(1 - \lambda) y_j + \lambda y_j^{(2)}] - \alpha^A \sum_{j=1}^n (\mu_{ij} - p_{0ij})^2 \quad \text{s.t.} \quad \sum_{j=1}^n \mu_{ij} = 1, \quad (5)$$

where $y_1^{(2)} = \bar{s}$ and $y_j^{(2)} = (Y_2 - \bar{s})/(n - 1)$ for $j > 1$ denote Period-2 income for each rank.¹⁶

F.2 Optimal belief under anticipated Period 2

Applying the Lagrangian to (5) as in Section 2, the first-order condition for each position j is

$$(1 - \lambda) y_j + \lambda y_j^{(2)} - 2\alpha^A (\mu_{ij} - p_{0ij}) = \nu,$$

where ν is the multiplier on the adding-up constraint. Summing over j and using $\sum_j \mu_{ij} = \sum_j p_{0ij} = 1$, together with $\sum_j y_j = n\bar{y}$ and $\sum_j y_j^{(2)} = Y_2$, yields $\nu = (1 - \lambda)\bar{y} + \lambda Y_2/n$. The

¹⁶The cost coefficient α^A under anticipated Period 2 may differ from its Baseline value because subjects internalize the realized utility loss from optimizing the allocation under a distorted belief. The robustness results below hold for any $\alpha^A > 0$.

optimal distortion for position j is

$$\mu_{ij}^* - p_{0ij} = \frac{(1 - \lambda)(y_j - \bar{y}) + \lambda(y_j^{(2)} - Y_2/n)}{2\alpha^A}. \quad (6)$$

This generalizes the baseline distortion in Equation (3) of Section 2: the first term in the numerator is the Period-1 income premium relative to the mean, reweighted by the probability that Period 1 is the paid period; the second is the analogous Period-2 income premium.

Symmetric equilibrium. The Period-2 premium depends on the implemented allocation $\bar{s}$, which itself depends on the population's beliefs. Under random dictator, $\bar{s}$ equals a uniformly drawn subject's choice; in the symmetric equilibrium where all subjects share Rank-1 belief μ_R , this gives $\bar{s} = \mu_R Y_2$. Substituting into (6) for rank 1 and using $y_1^{(2)} = \bar{s} = \mu_R Y_2$ together with the uniform prior $p_{0i1} = 1/n$:

$$\mu_R^* - \frac{1}{n} = \frac{(1 - \lambda)(y_R - \bar{y}) + \lambda Y_2(\mu_R^* - 1/n)}{2\alpha^A}. \quad (7)$$

Solving the fixed point:

$$\mu_R^* - \frac{1}{n} = \frac{(1 - \lambda)(y_R - \bar{y})}{2\alpha^A - \lambda Y_2}. \quad (8)$$

For the symmetric equilibrium to be interior, the denominator must be positive: $2\alpha^A > \lambda Y_2$.

F.3 Persistence of overconfidence and the inequality effect

The central predictions of the baseline model survive the extension.

Proposition 1 (Persistence of overconfidence). *For any $\lambda \in (0, 1)$ and $y_R > \bar{y}$, the symmetric equilibrium under anticipated Period 2 has $\mu_R^* > 1/n$. Subjects remain strictly overconfident about being Rank 1.*

Proof. The numerator of (8), $(1 - \lambda)(y_R - \bar{y})$, is strictly positive; the denominator is positive by the interior-equilibrium condition. Therefore $\mu_R^* - 1/n > 0$. The underlying logic is identical to that of Section 2: at $\mu_R = 1/n$, the marginal anticipatory benefit of raising μ_R is $(1 - \lambda)(y_R - \bar{y}) > 0$ (first order), while the marginal decision cost is $2\alpha^A(\mu_R - 1/n) = 0$ (second order). The asymmetry from Brunnermeier and Parker (2005) that drives the baseline result is unchanged. $\square$

Proposition 2 (Inequality-overconfidence link). *Holding total Period-1 income Y and Period-2 parameters fixed,*

$$\frac{\partial(\mu_R^* - 1/n)}{\partial \pi} = \frac{(1 - \lambda)Y}{2\alpha^A - \lambda Y_2} > 0,$$

so higher Period-1 inequality raises the equilibrium Rank-1 belief. Moreover, the implemented Period-2 allocation $\bar{s} = \mu_R^ Y_2$ also rises in π : higher Period-1 inequality propagates to a more unequal Period-2 implementation through the belief channel.*

Proof. Substituting $y_R - \bar{y} = (\pi - 1/n)Y$ into (8) and differentiating with respect to π gives the first claim; $\bar{s} = \mu_R^* Y_2$ in symmetric equilibrium gives the second. $\square$

The two main predictions of the baseline model therefore extend unchanged. Subjects exhibit strict overconfidence about Rank 1, and higher Period-1 inequality raises both the equilibrium belief and the implemented Period-2 allocation. Whether one assumes that subjects form beliefs in ignorance of downstream decisions (the Baseline) or in full awareness of them, the same qualitative pattern emerges, with the same comparative-statics signature in π .